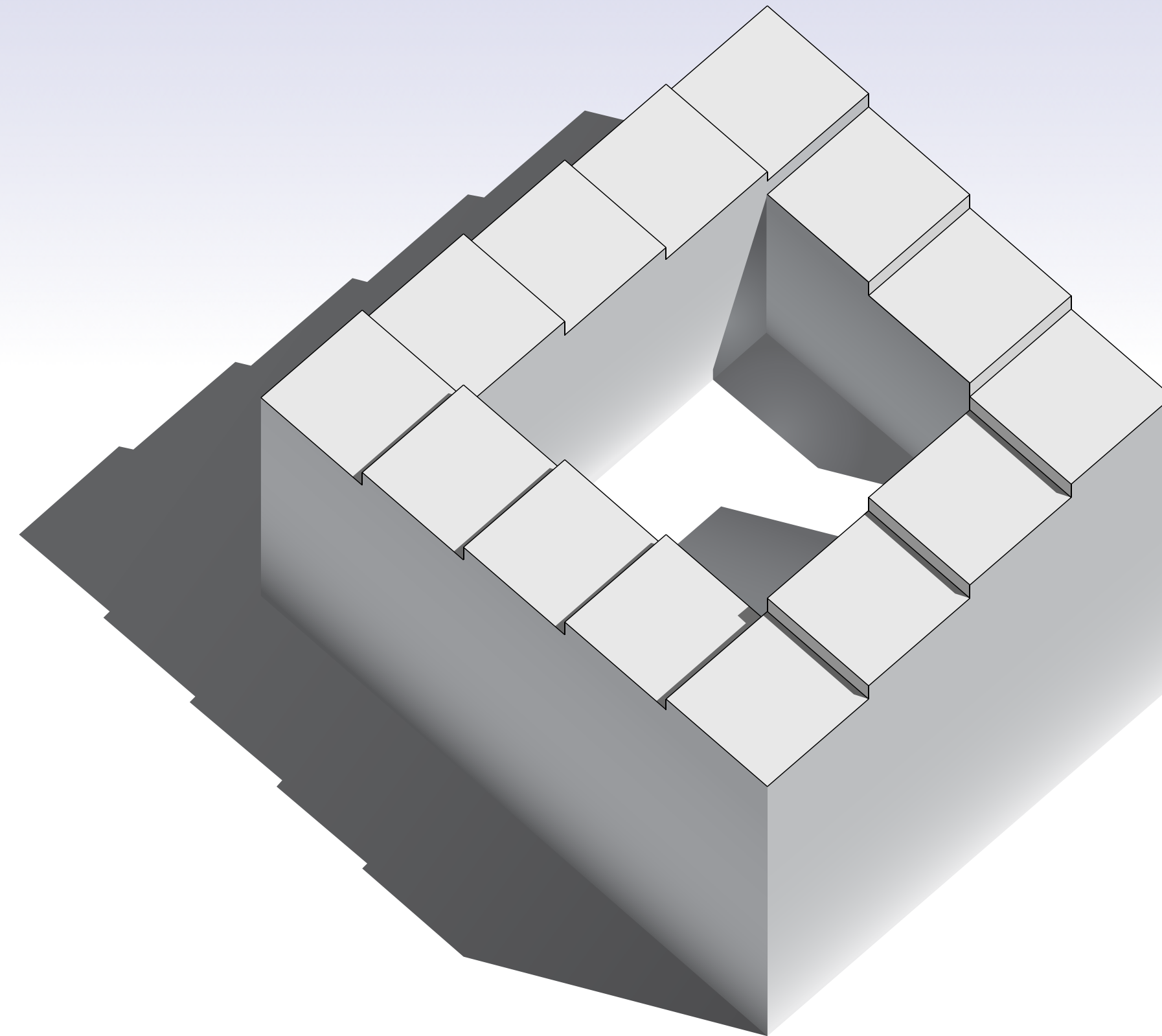
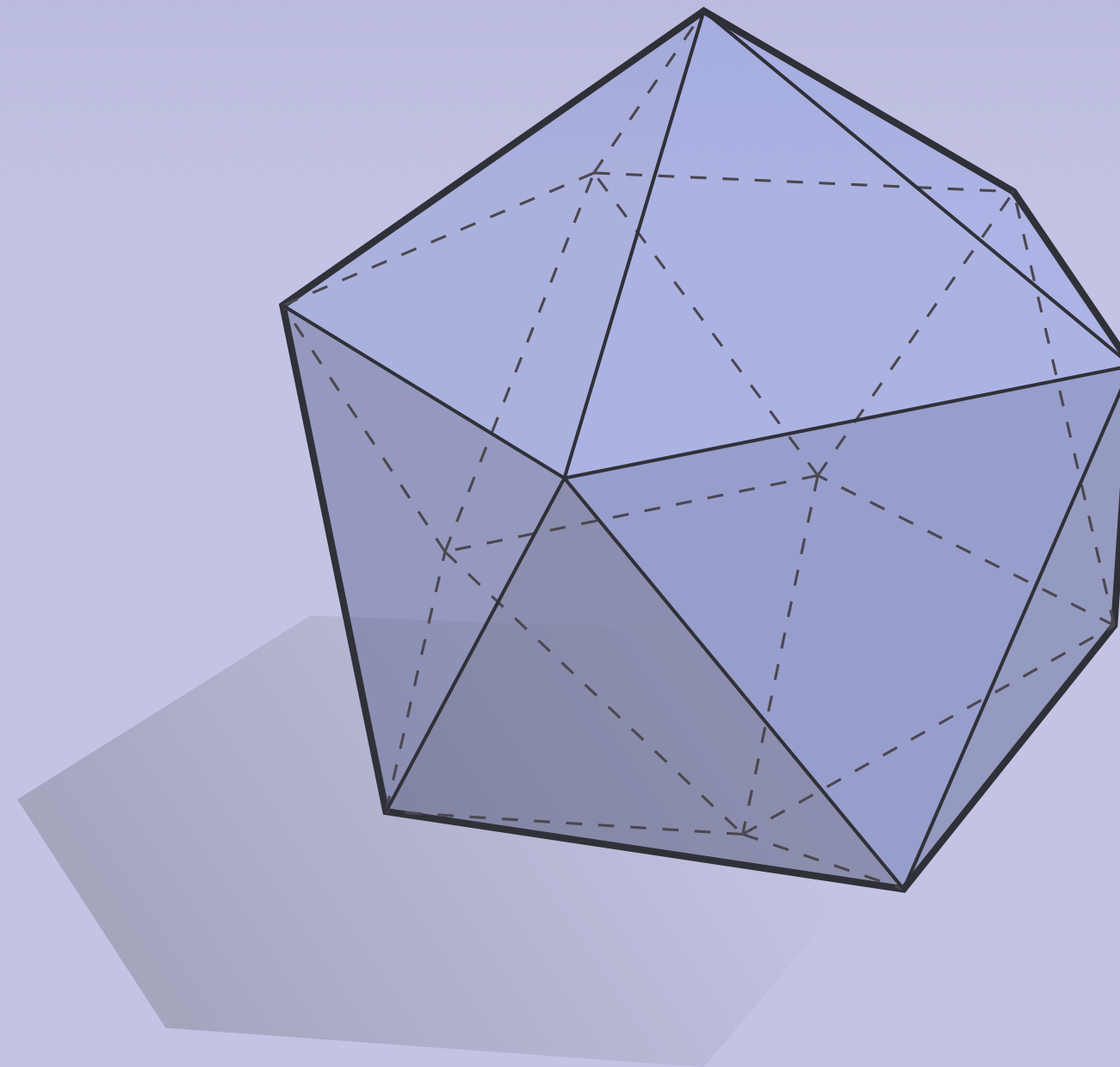




DISCRETE DIFFERENTIAL
GEOMETRY:
AN APPLIED INTRODUCTION
Keenan Crane • CMU 15-458/858

GEODESICS II



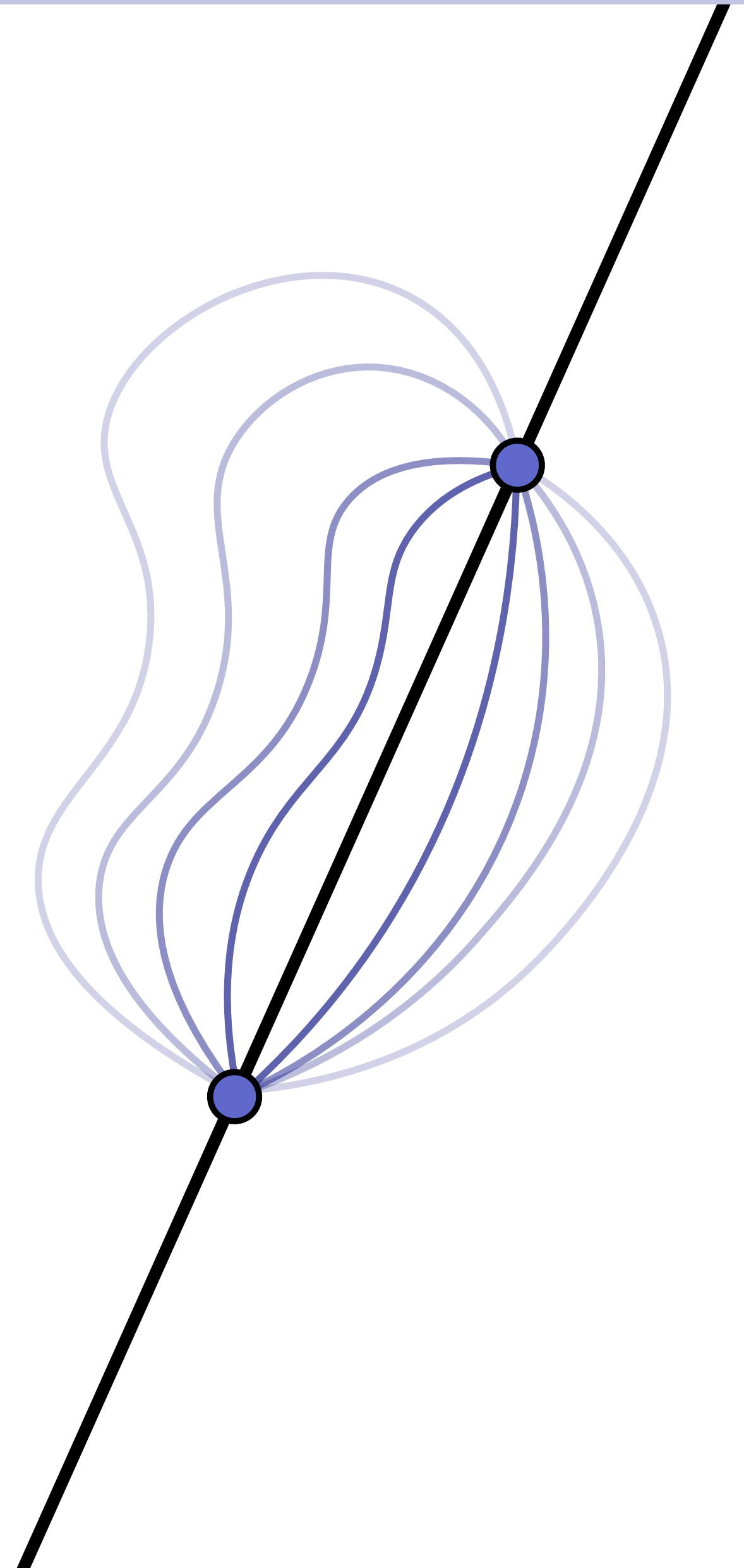
DISCRETE DIFFERENTIAL
GEOMETRY:

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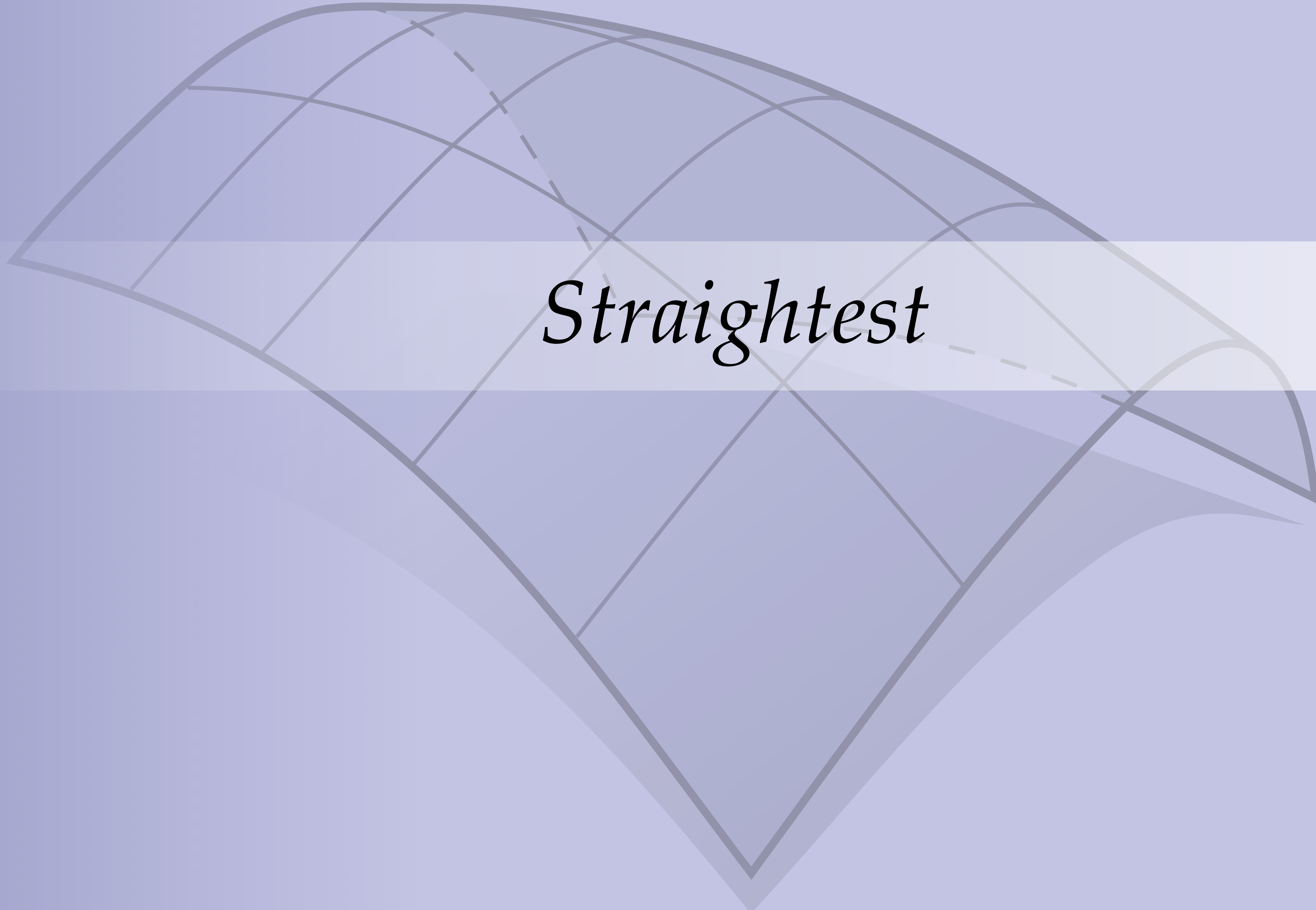
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Geodesics—Recap

- Generalize the notion of *lines* to curved spaces
- Ordinary lines have two basic features:
 1. straightest — no curvature / acceleration
 2. shortest — (locally) minimize length
- Geodesics share these same *local* properties, but may exhibit different behavior *globally*
- Part of the “origin story” of both classical and differential geometry...



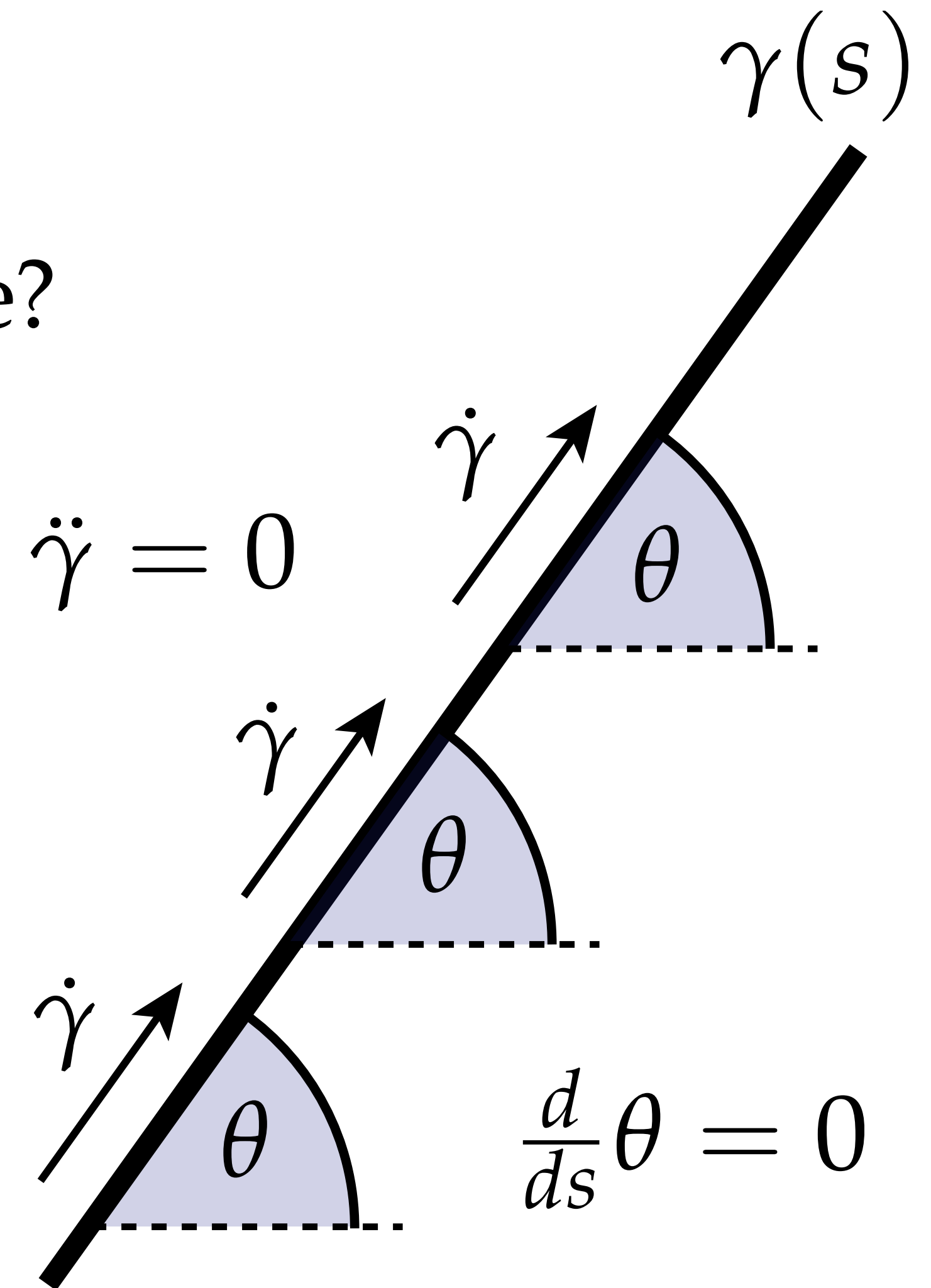
Key idea: geodesic is straightest, (locally) shortest curve



Straighttest

Straightest Paths

- A Euclidean line can be characterized as a curve that is “as straight as possible”
- **Q:** How can we make this statement more precise?
 - **geometrically:** no curvature
 - **dynamically:** no acceleration
- How can we generalize to curves in manifolds?
 - **geometrically:** no *geodesic curvature*
 - **dynamically:** zero *covariant derivative*



Straightness — Geometric Perspective

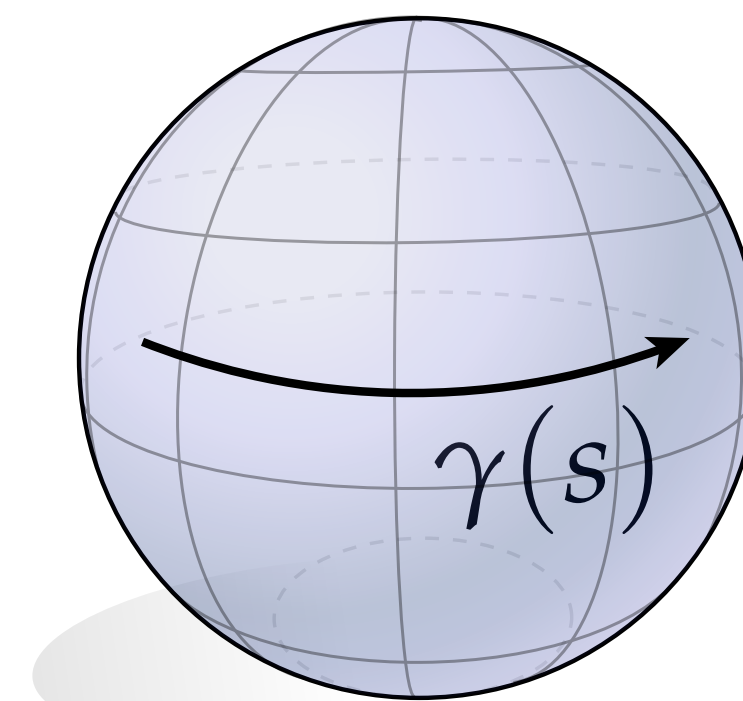
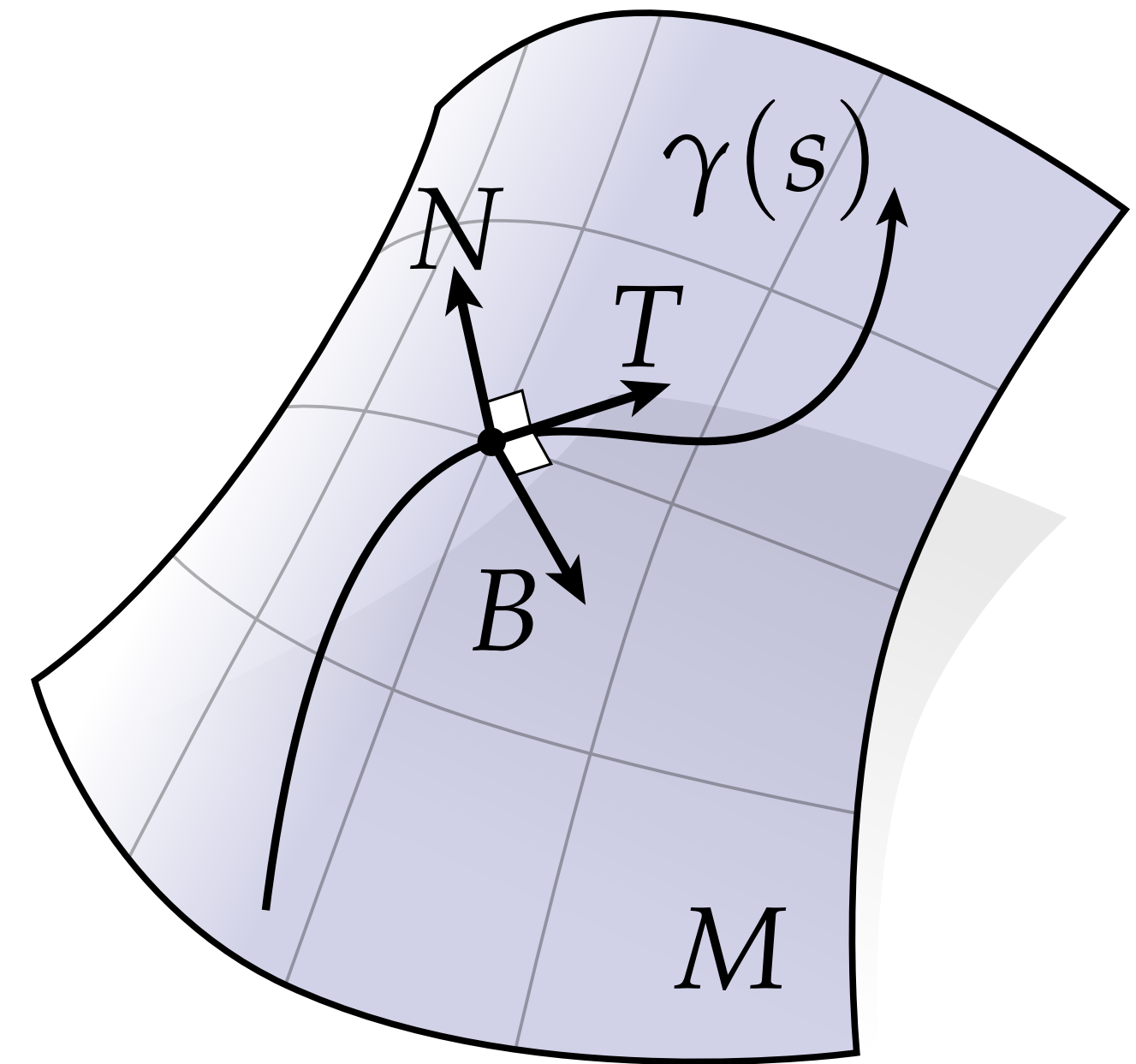
- Consider a curve $\gamma(s)$ with tangent T in a surface with normal N , and let $B := T \times N$.
- Can decompose “bending” into two pieces:

$$\kappa_n := \left\langle N, \frac{d}{ds} T \right\rangle \quad \text{normal curvature}$$

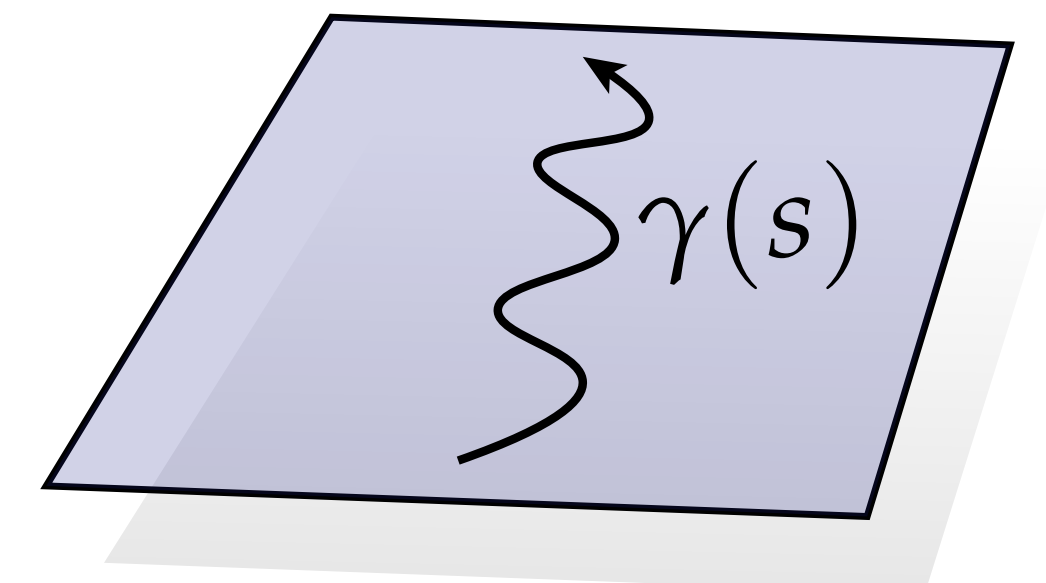
$$\kappa_g := \left\langle B, \frac{d}{ds} T \right\rangle \quad \text{geodesic curvature}$$

- Curve is “forced” to have normal curvature due to curvature of M
- Any additional bending beyond this minimal amount is geodesic curvature

Key idea: geodesic is curve where $\kappa_g = 0$



large κ_n ;
small κ_g

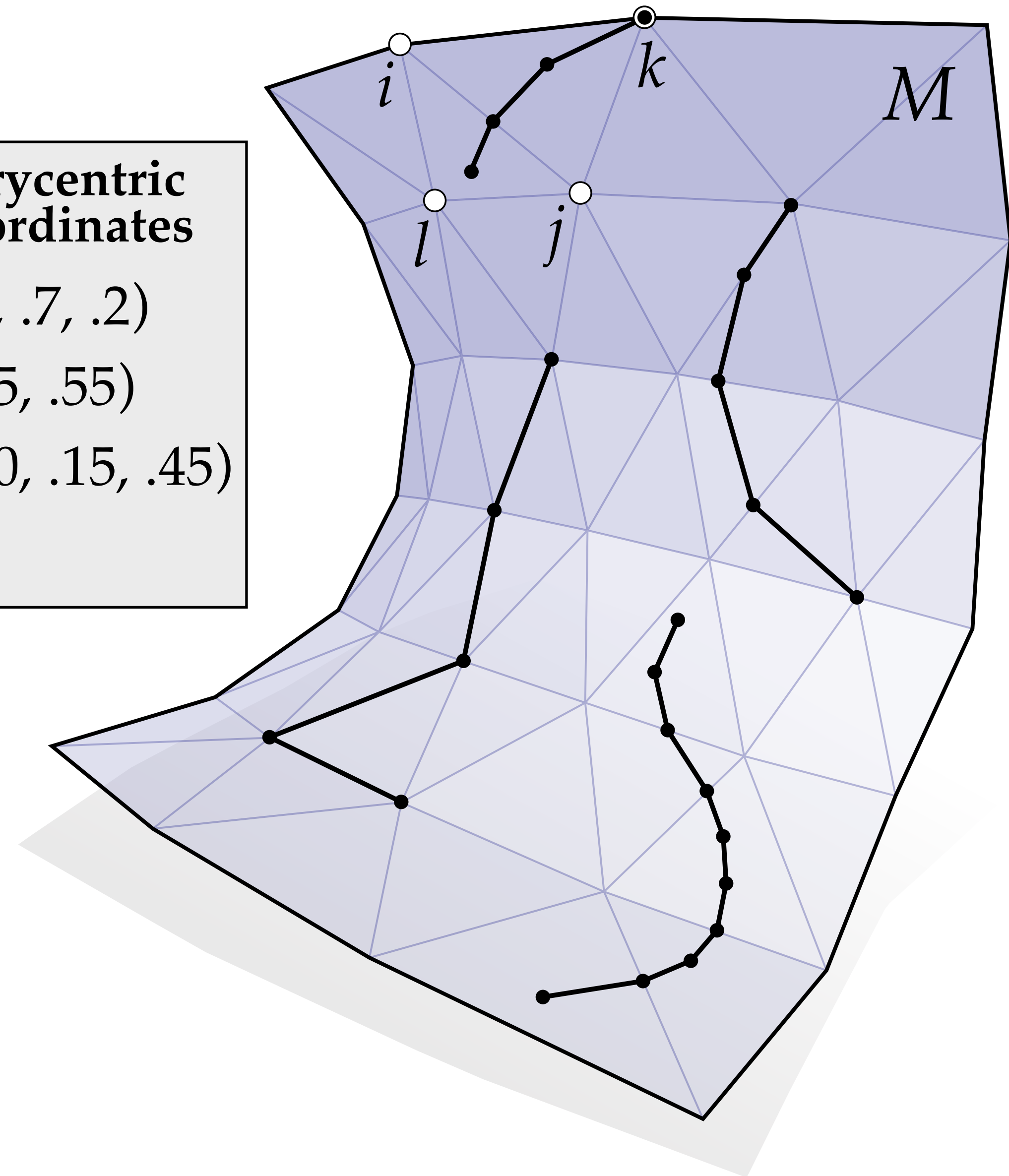


large κ_g ;
small κ_n

Discrete Curves on Discrete Surfaces

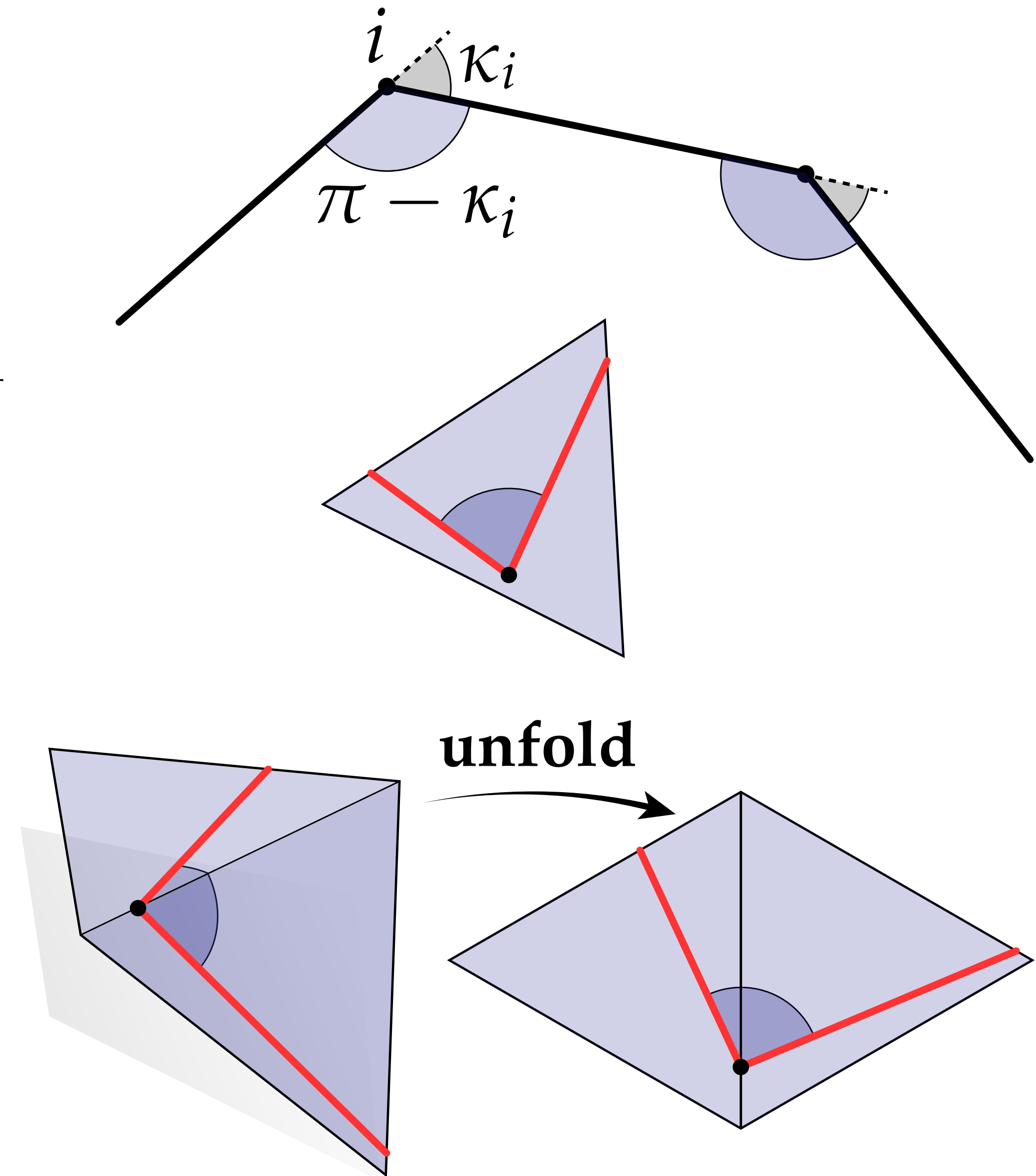
- To understand straightest curves on discrete surfaces, first have to define what we mean by a *discrete curve*
- One definition: a discrete curve in a simplicial surface M is any continuous curve γ that is piecewise linear in each simplex
- Doesn't have to be a path of edges: could pass through faces, have multiple vertices in one face, ...
- Encode as sequence of simplices (not all same degree), and barycentric coordinates for each simplex

simplex	barycentric coordinates
ilj	(.1, .7, .2)
ij	(.45, .55)
ijk	(.40, .15, .45)
k	(1)



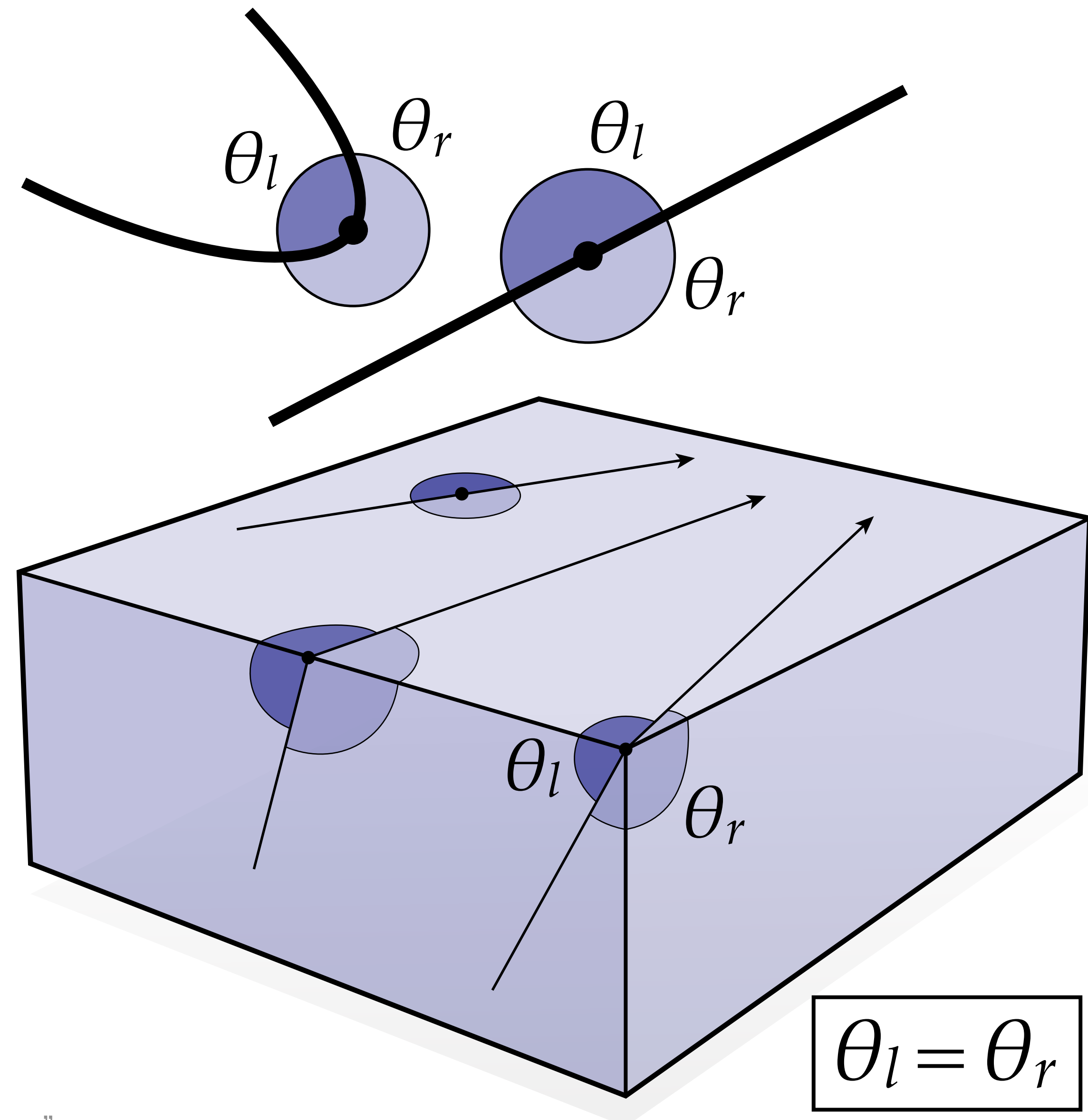
Discrete Geodesic Curvature

- For planar curve, one definition of discrete curvature was *turning angle* κ_i
- Since most points of a simplicial surface are *intrinsically* flat, can adopt this same definition for discrete geodesic curvature
- *Faces*: just measure angle between segments
- *Edges*: “unfold” and measure angle
- *Vertices*: not as simple—can’t unfold!
 - Recall trouble w/ **shortest** geodesics...



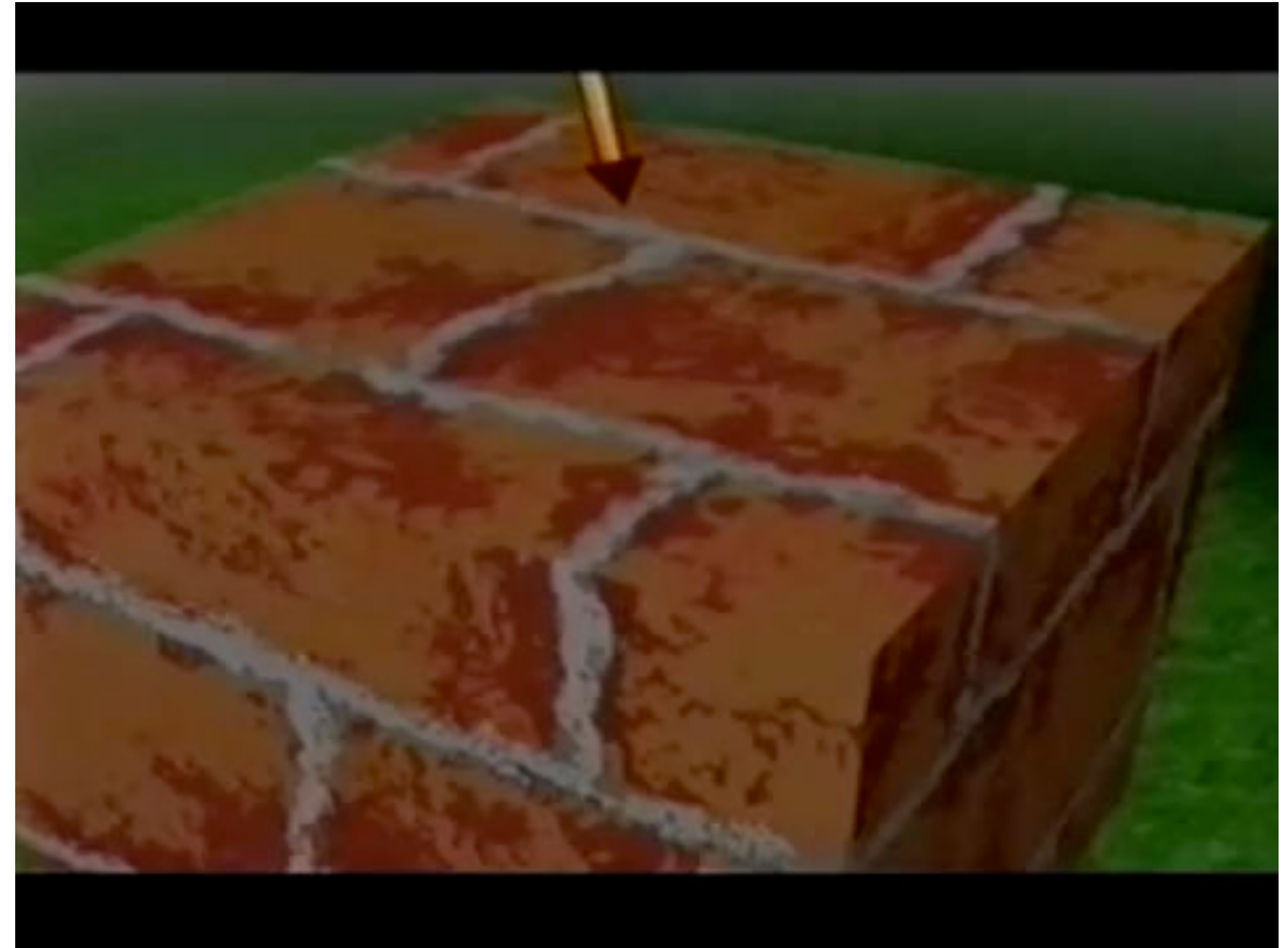
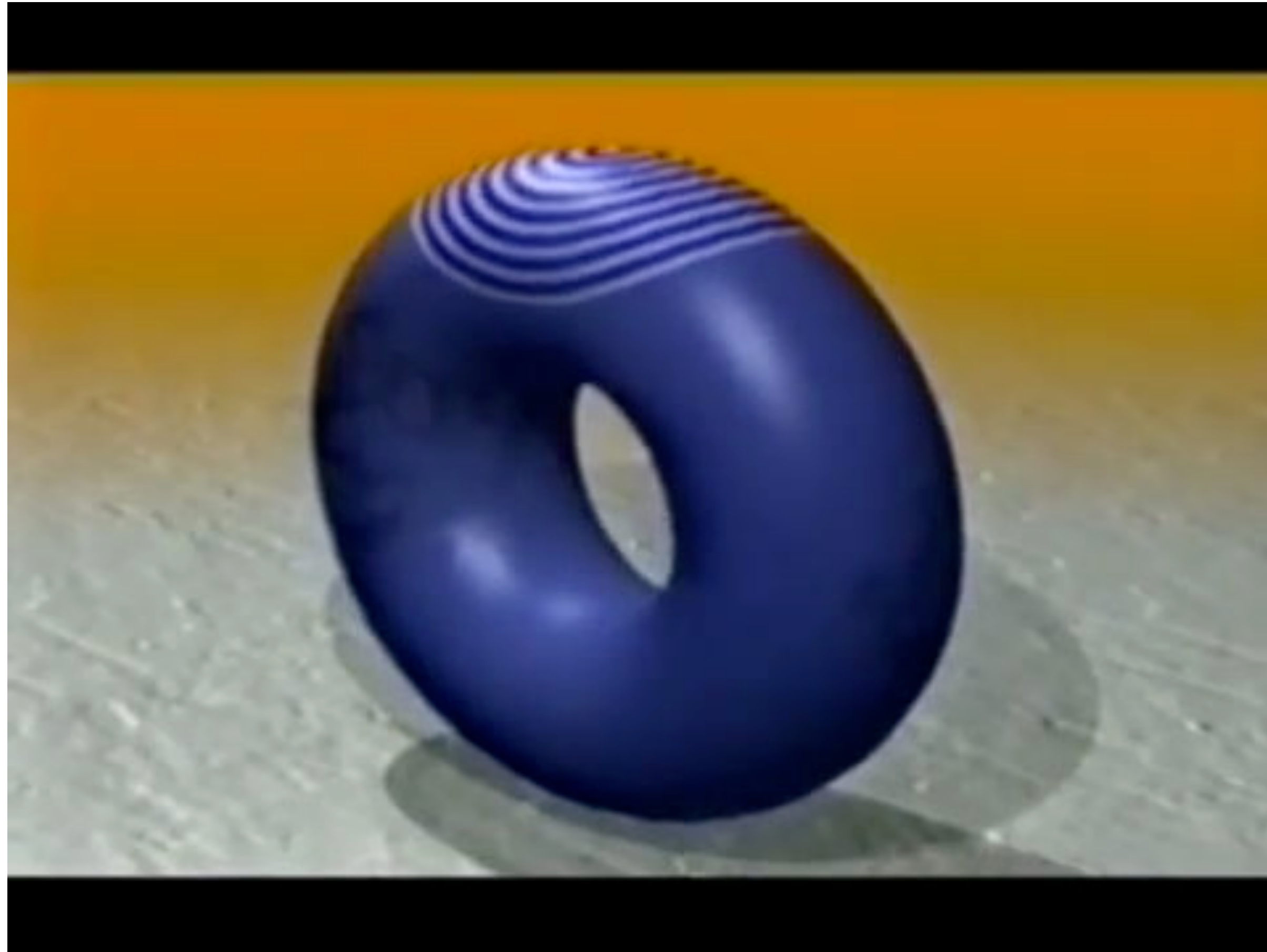
Discrete Straightest Geodesics

- In the smooth setting, characterized geodesics as curves with zero geodesic curvature
- In the discrete setting, have a hard time at vertices: can't unfold, no *shortest* paths through some vertices...
- Alternative smooth characterization: just have same angle on either side of the curve
- Translates naturally to the discrete setting: equal angle sum on either side of the curve
- Provides definition of discrete **straightest** geodesics (Polthier & Schmies 1998)



Geodesics and Waves

Might seem that geodesics are “unlikely” to pass exactly through a vertex, but consider simulating a continuous wavefront—how should it behave when it hits a vertex?



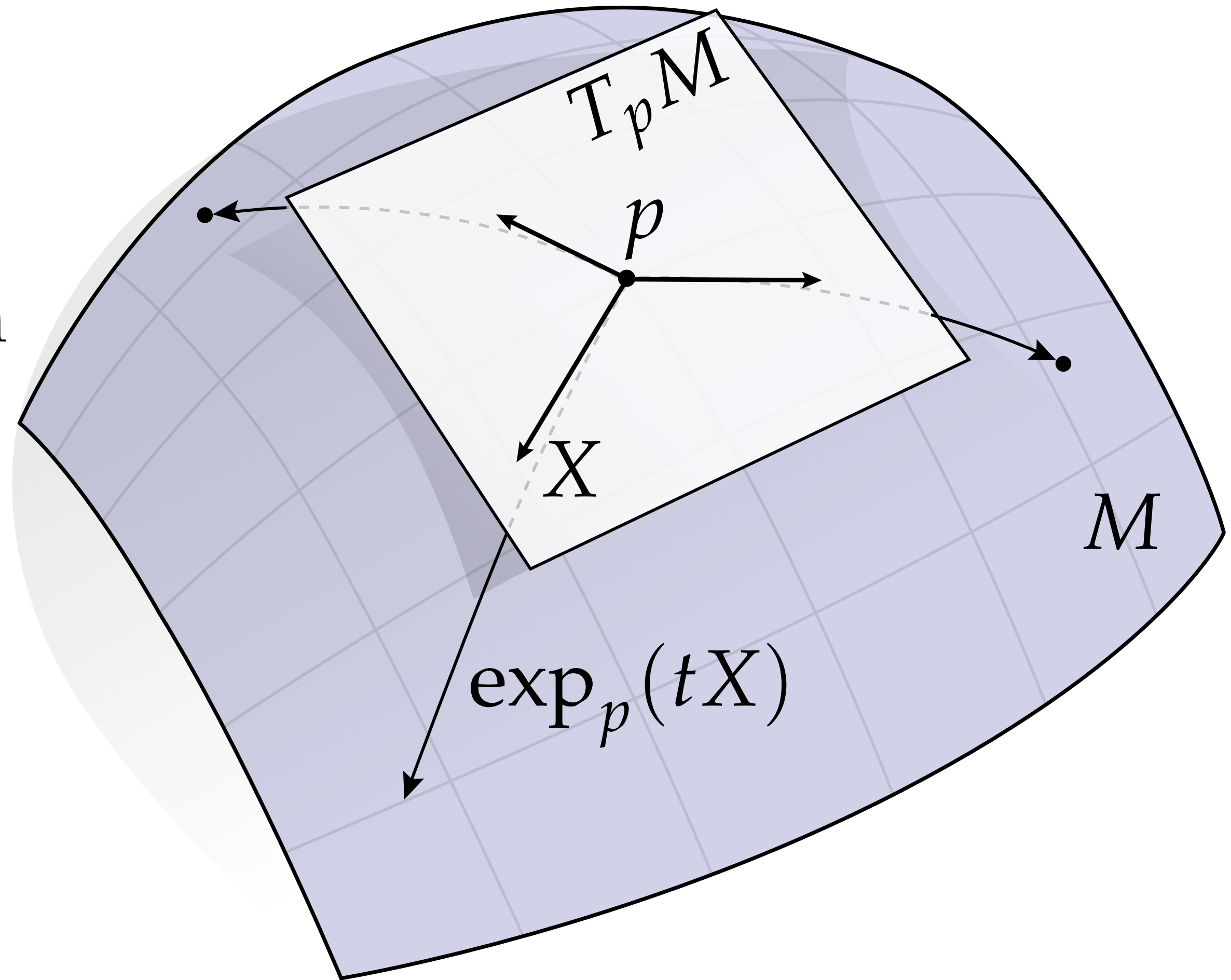
Exponential Map

- At a point p of a smooth surface M , the *exponential map* $\exp_p: T_pM \rightarrow M$ takes a tangent vector X to the point reached by walking along a geodesic in the direction $X/|X|$ for distance $|X|$

exponential map at p tangent vectors points

$$\exp_p: T_pM \rightarrow M$$

- Can also imagine that $\exp$ “wraps” the tangent plane around the surface

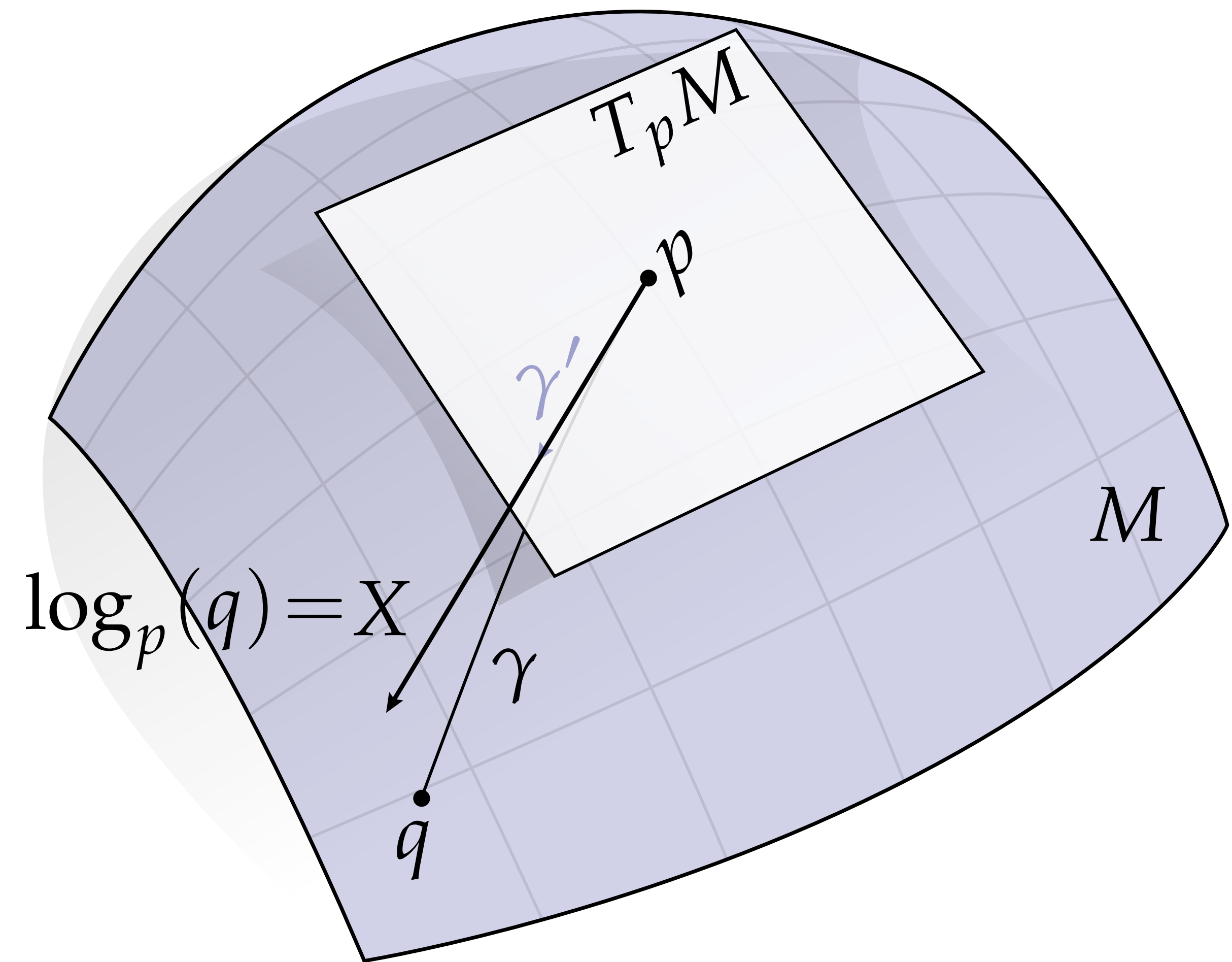


Key idea: provides notion of “translation” for curved domains

Logarithmic Map

- **Q:** Is the exponential map *surjective*? I.e., can we reach every point q from p ?
- **A:** Yes (Hopf-Rinow): Consider a smooth surface M without boundary. Then
 - find the shortest geodesic γ from p to q
 - let X be a vector in direction γ' w/ length $|\gamma|$
 - then by construction, $\exp_p(X) = q$
 - Can also write $\log_p(q) = X$
- Map from q to X is called the *log map*

Q: Is the log map uniquely determined?



log map
at p

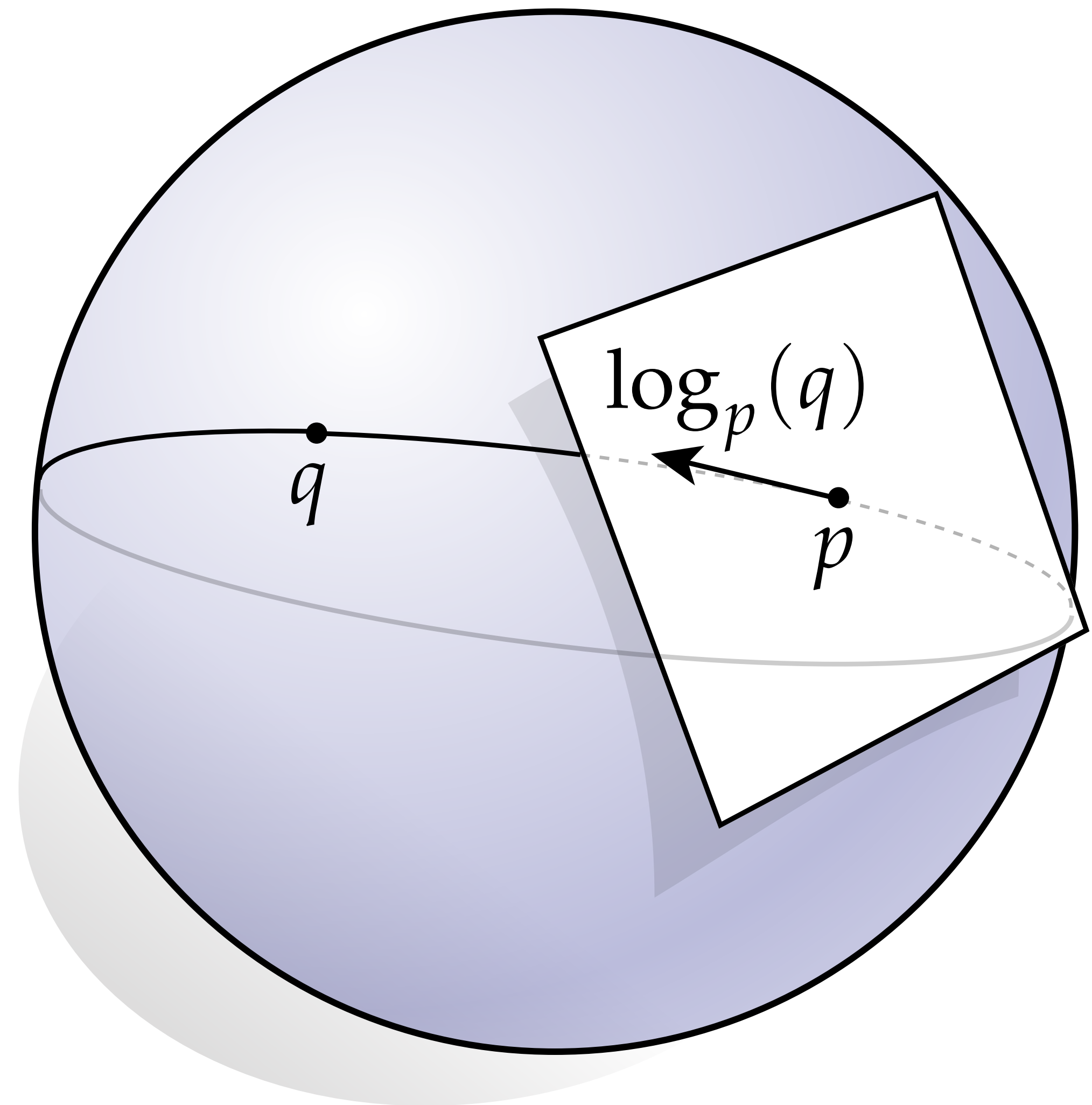
points

tangent
vectors

$$\log_p : M \rightarrow T_p M$$

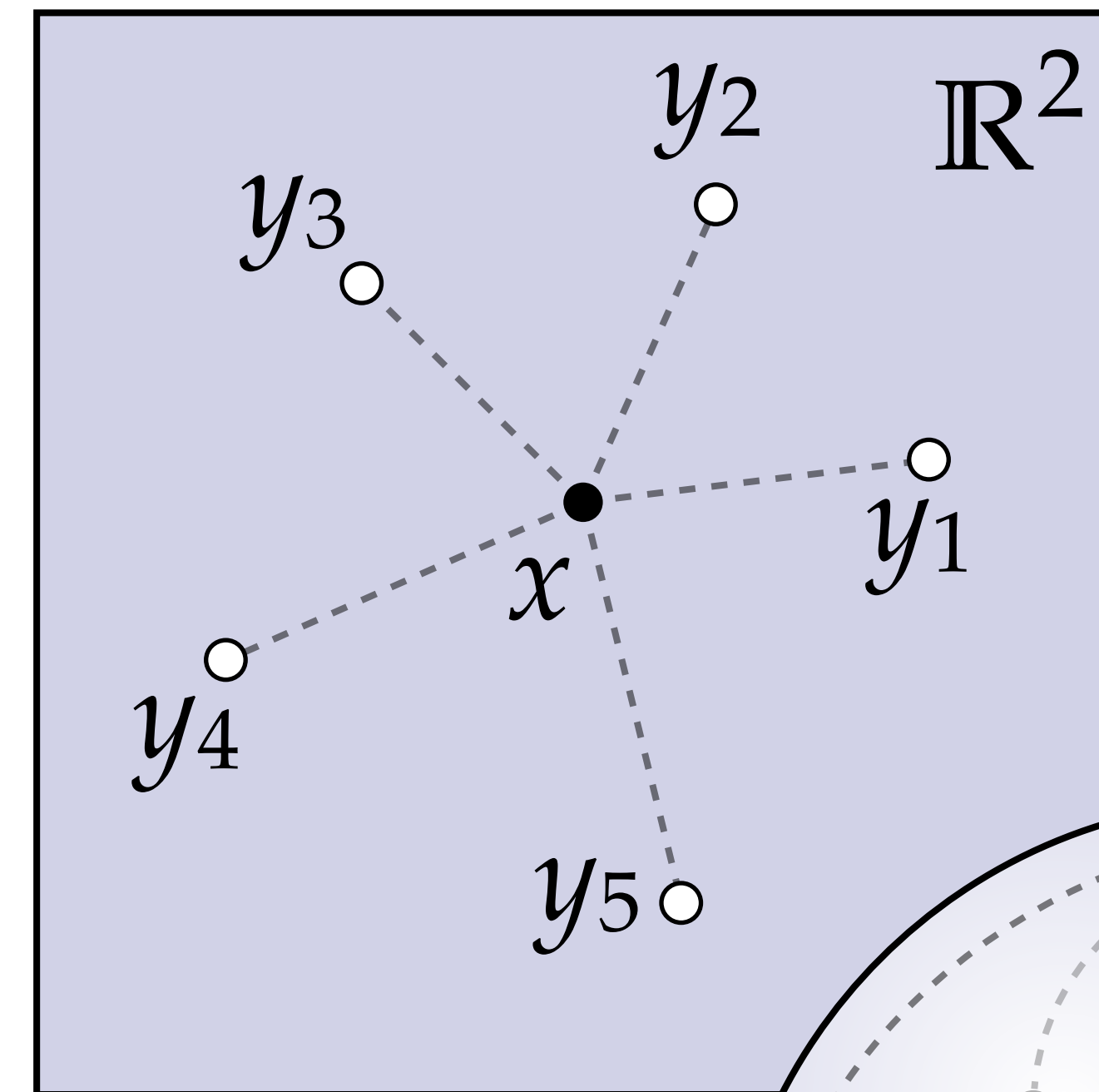
Exponential Map—Injectivity

- Equivalently, is the exponential map always *injective*? (I.e., is there a unique geodesic that takes us from p to q ?)
- No! Consider the exponential map on the sphere...
- By convention, log map therefore gives the *smallest* vector X such that $\exp_p(X) = q$
- **Q:** Why are exp / log map useful?
- **A:** Allows us to *locally* work with points on curved spaces as though they are just vectors in a flat space



Averages on Surfaces

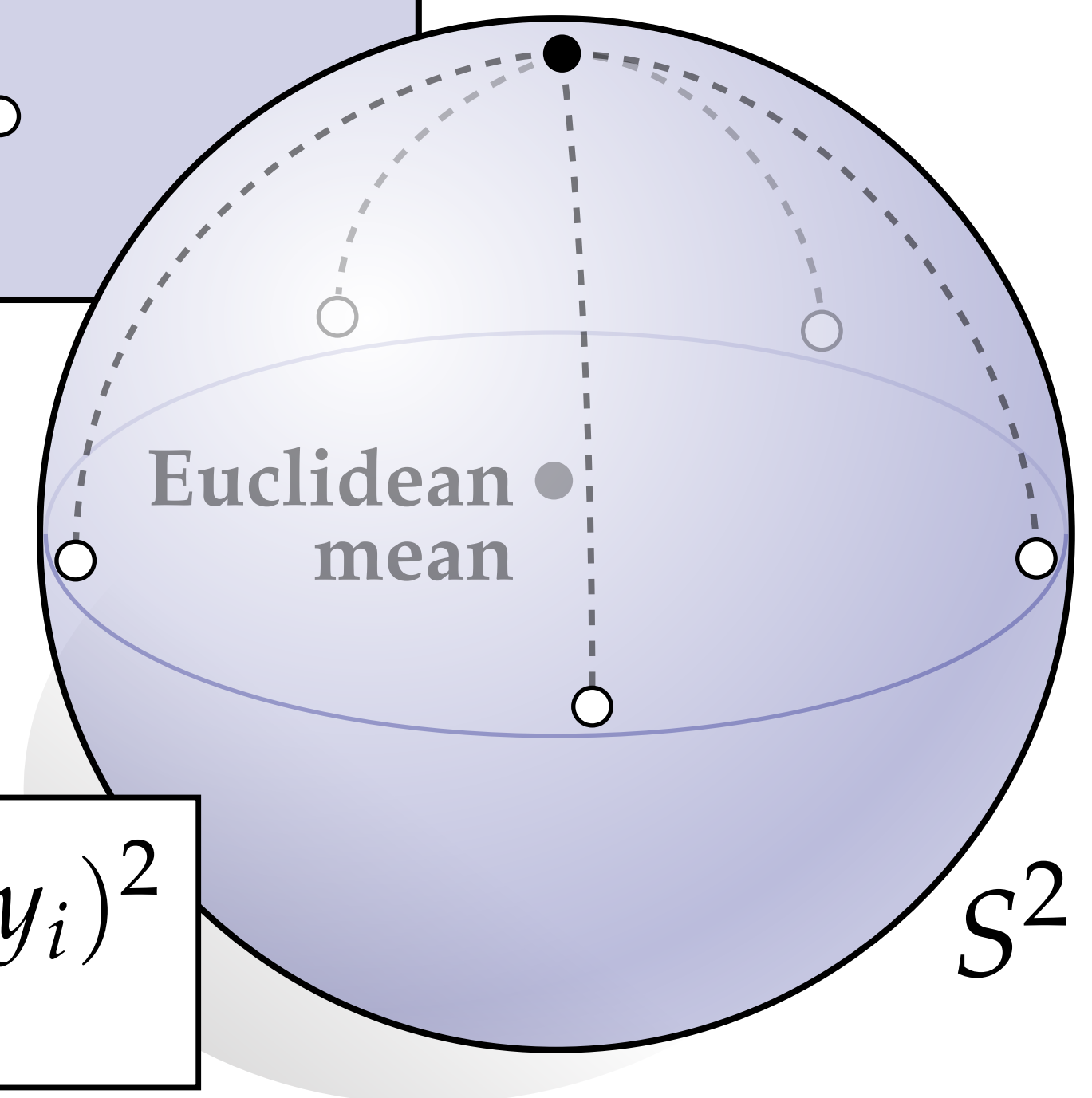
- Average of points in the plane is easy: just add up coordinates, divide by number of points
- How do we talk about an average of points on a curved surface?
 - average of coordinates may no longer be on the surface
 - might not even know how surface is embedded into space...
- Motivates idea of *Karcher mean*:
 - average is point that minimizes sum of squared geodesic distances to all points
 - in the plane, agrees with usual notion of “average” in the plane (why?)



Euclidean mean

$$x = \frac{1}{n} \sum_i y_i$$

Karcher mean



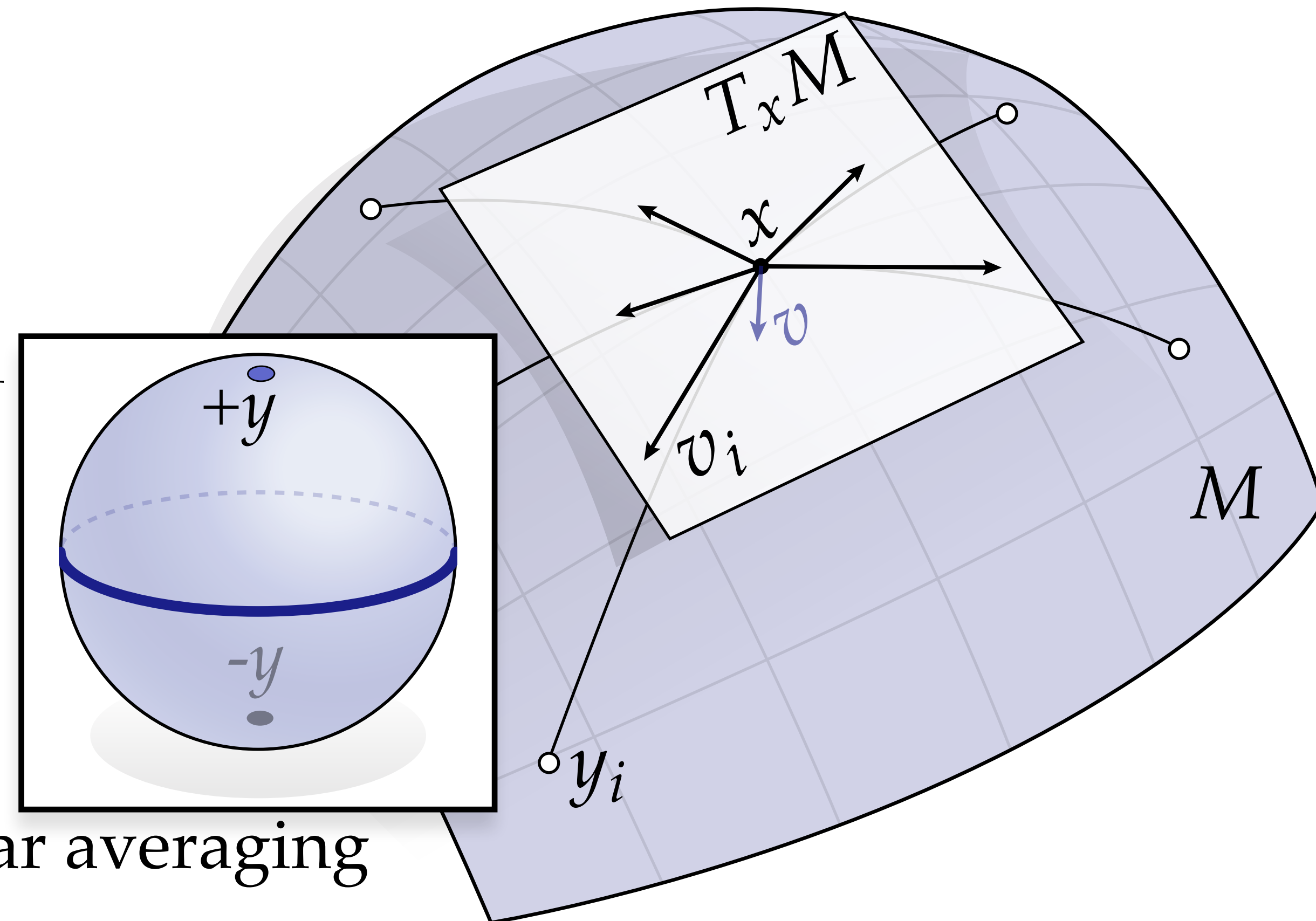
$$\min_x \frac{1}{n} \sum_i d(x, y_i)^2$$

Karcher Mean via Log Map

- Want to compute mean of points y_i
- Iterative algorithm:
 - pick a random initial starting point x
 - compute the log v_i of all points y_i
 - compute the mean v of all the vectors v_i
 - move x to $\exp_x(v)$ and repeat
- Will quickly converge to *some* Karcher mean
 - in general may not be unique—consider two points $y_1 = -y_2$ on the sphere
- Can also be used to average, e.g., rotations

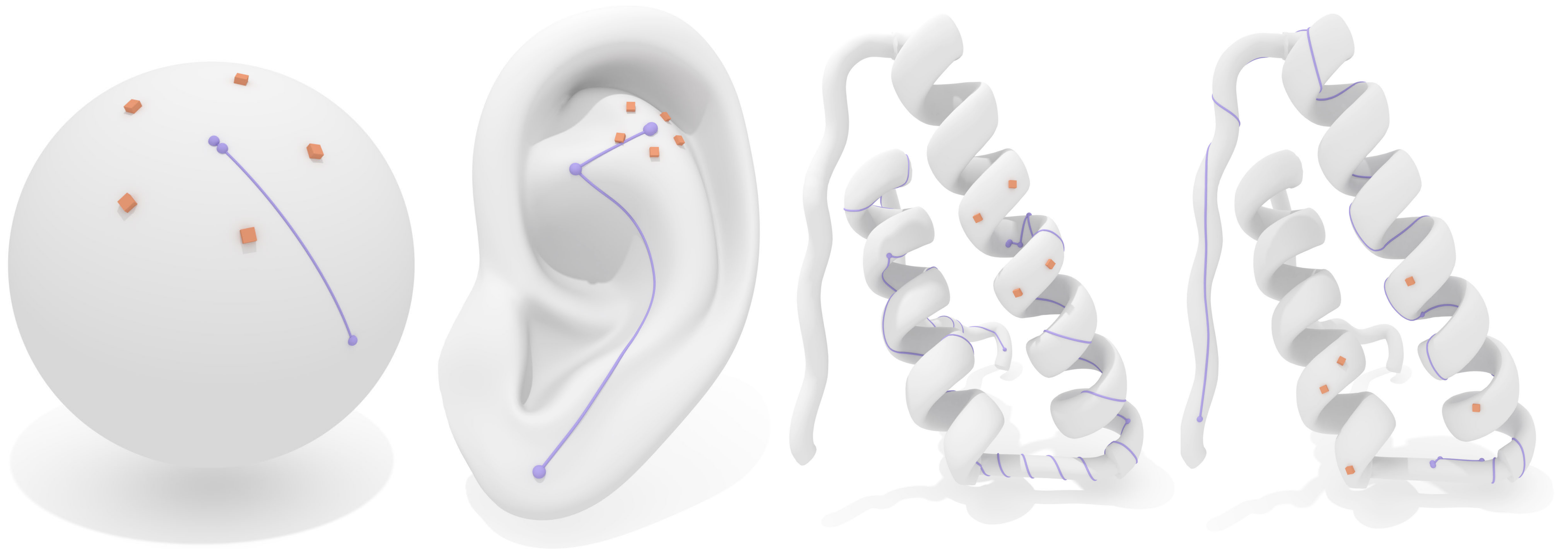
$$v \leftarrow \frac{1}{n} \sum_i \log_x(y_i)$$

$$x \leftarrow \exp_x(v)$$



Key idea: turn “curved averaging” into linear averaging

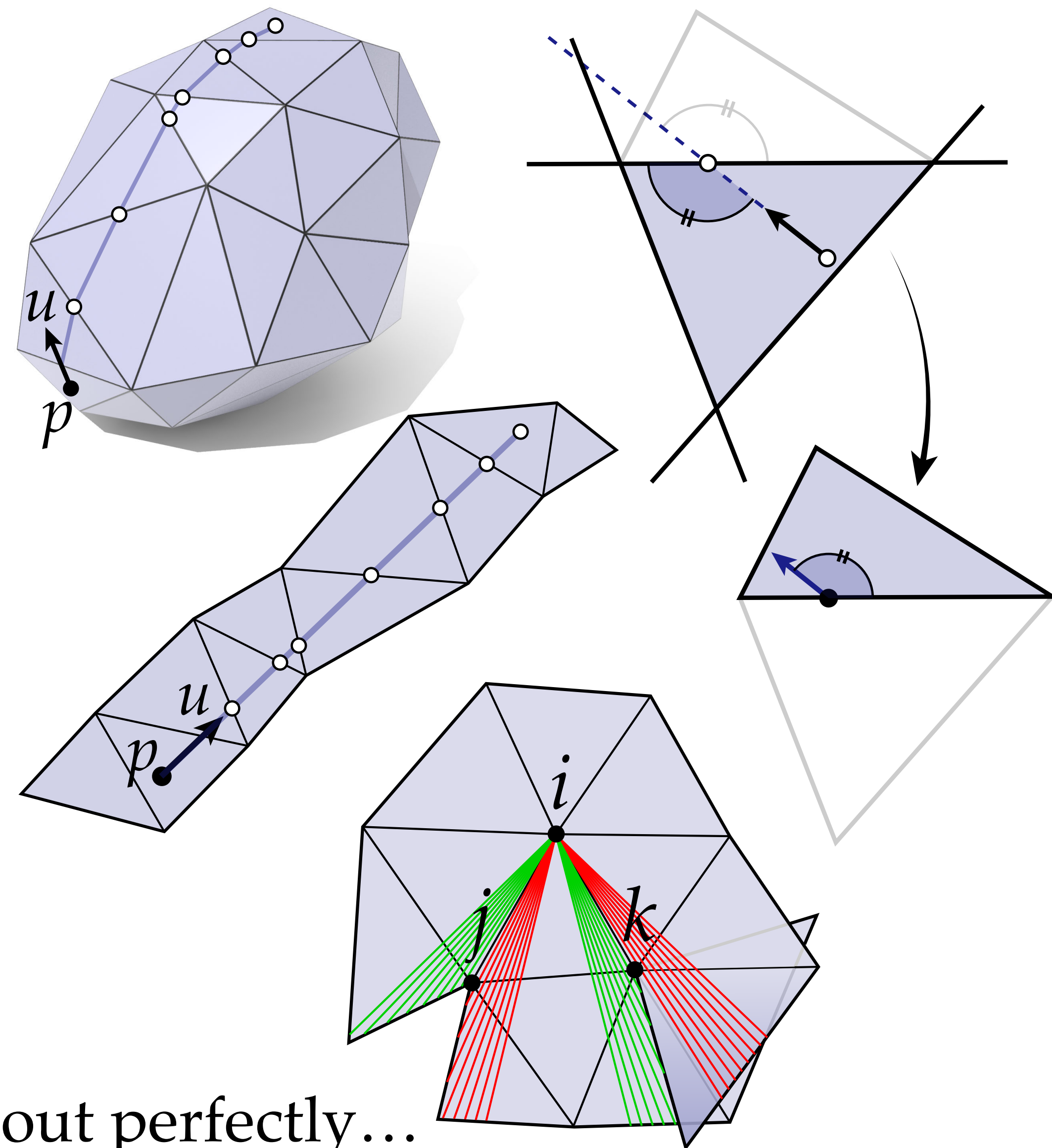
Karcher Mean — Examples



Notice: not always as easy as taking Euclidean average & projecting onto surface!

Discrete Exponential Map

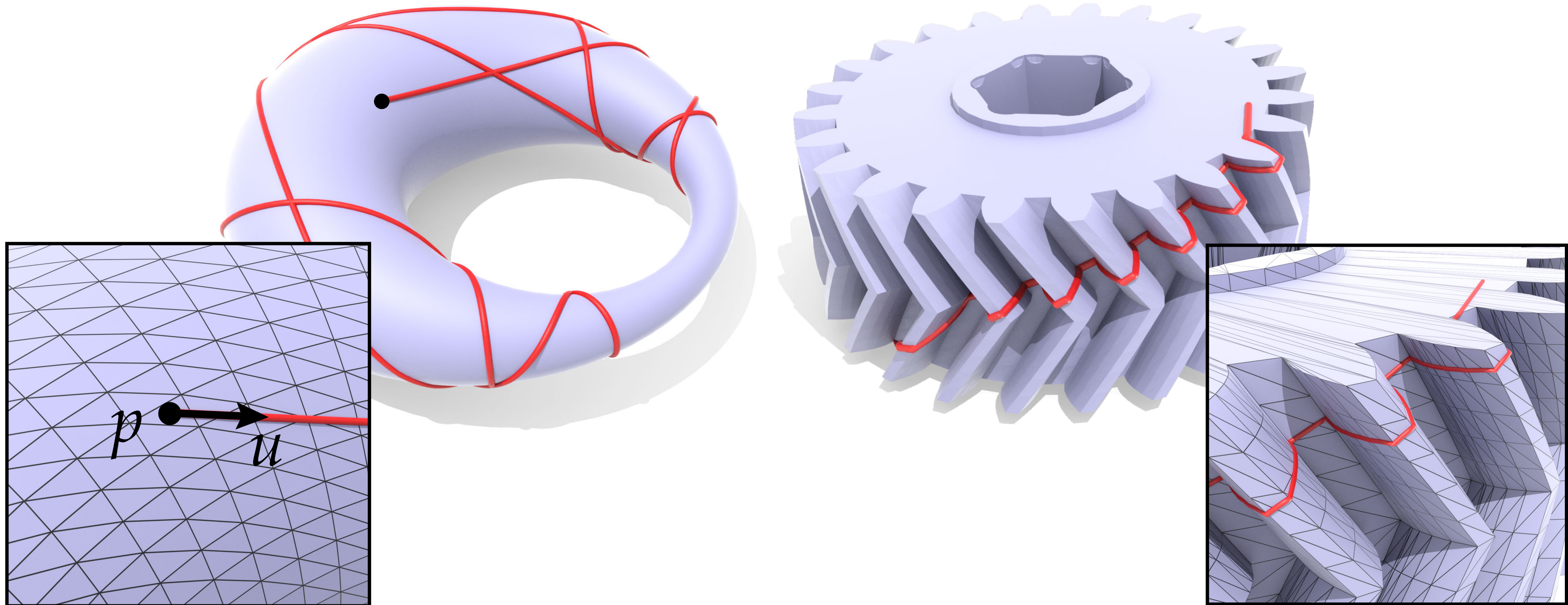
- Easy to evaluate exp map on discrete surfaces
- Given point p and vector u , start walking along u
 - *i.e.*, just intersect ray with edges of triangle
 - continue w/ same angle in next triangle
 - if we hit a vertex, continue in direction that makes equal angles (*straightest*)
- **Q:** How big is the injectivity radius?
- **A:** Distance to the closest cone vertex ($\Omega > 0$)
- **Q:** Is the discrete exponential map surjective?
- **A:** No! Consider a saddle vertex ($\Omega < 0$)



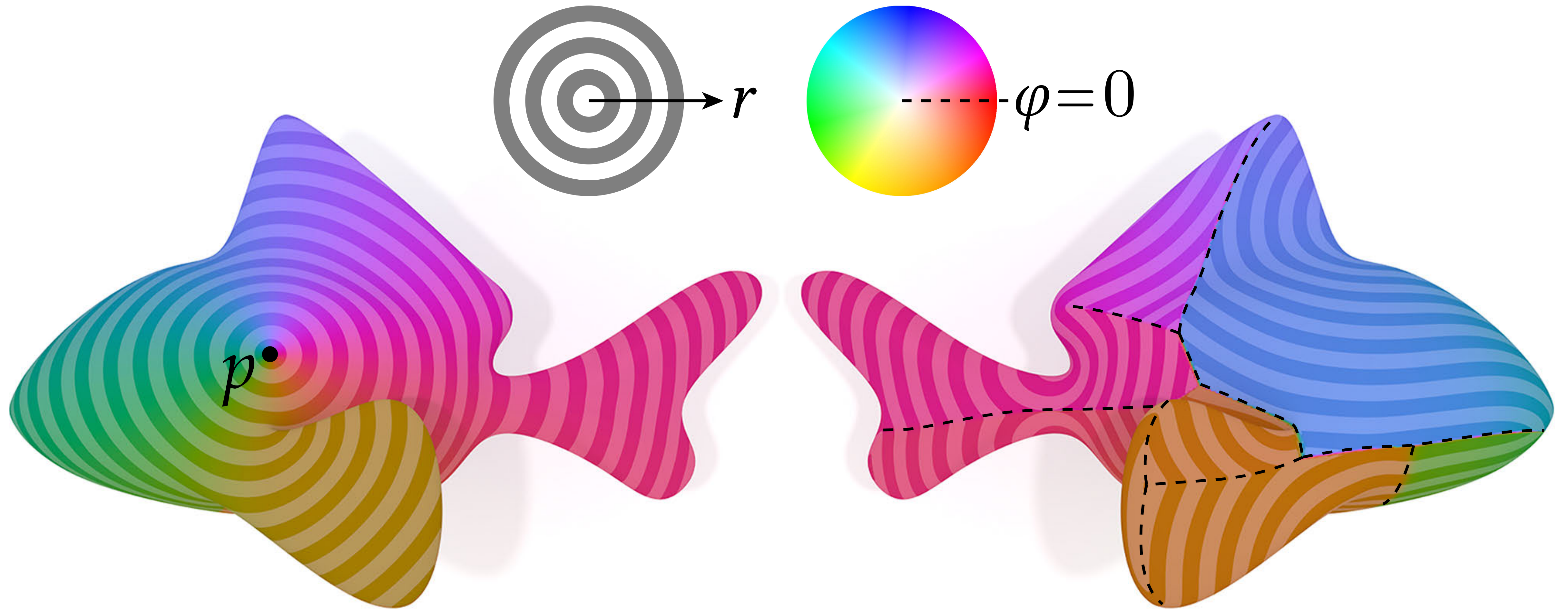
Notice: like “shortest”, “straightest” doesn’t work out perfectly...

Discrete Exponential Map—Examples

- Discrete exponential map provides a practical way to *approximate* geodesics on smooth surfaces (by triangulating them), and gives *exact* geodesics on discrete surfaces



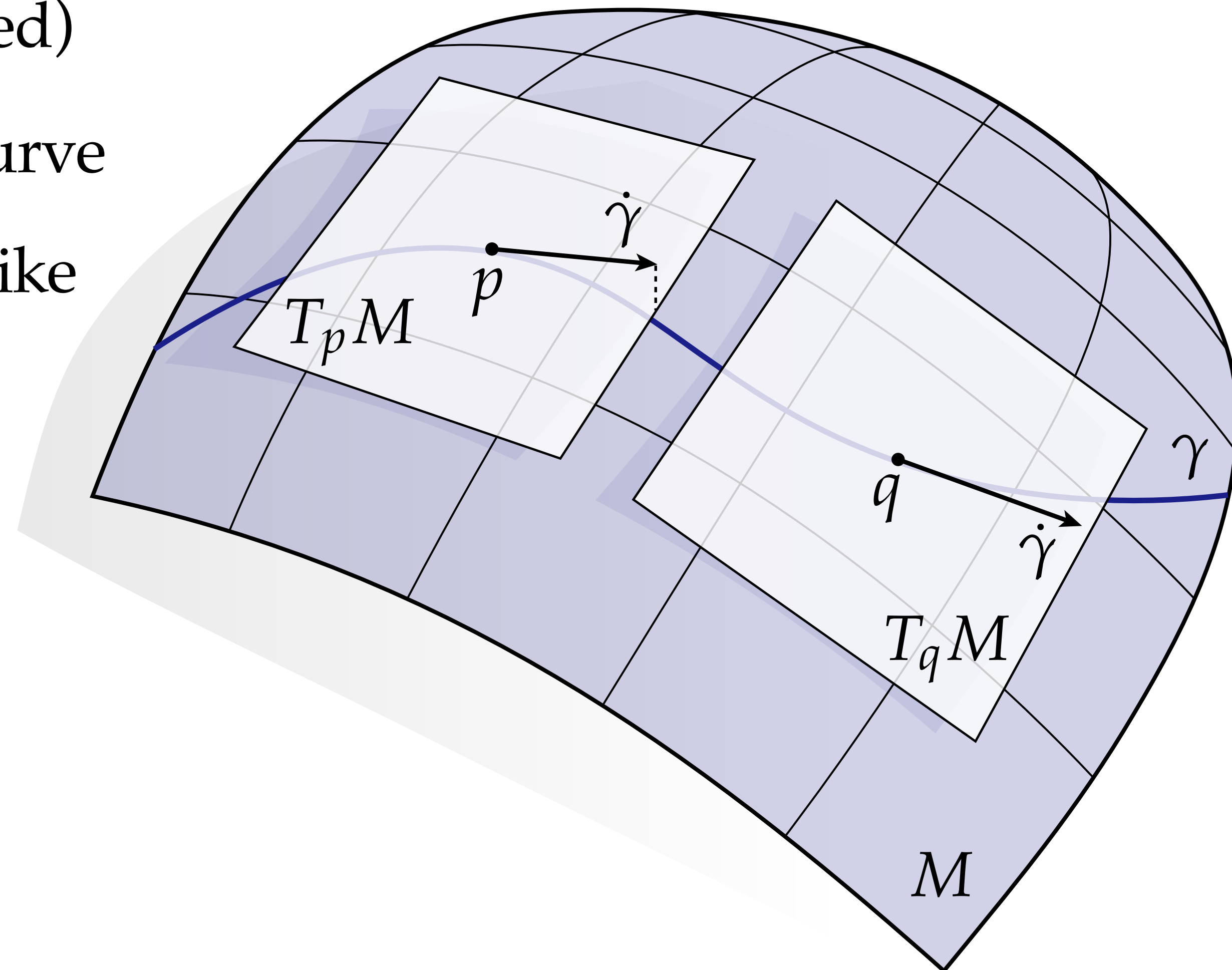
Computing the Log Map



Sharp, Soliman, Crane, "The Vector Heat Method" (2019)

Straightness—Dynamic Perspective

- Dynamic perspective: geodesic has zero tangential acceleration
- Consider curve $\gamma(t): [a,b] \rightarrow M$ (*not* unit speed)
- *Tangential velocity* is just the tangent to the curve
- *Tangential acceleration* should be something like the “tangential change in the tangent,” but:
 - **extrinsically**, change in tangent is not a tangent vector
 - **intrinsically**, tangents belong to different vector spaces
- So, how do we measure acceleration?



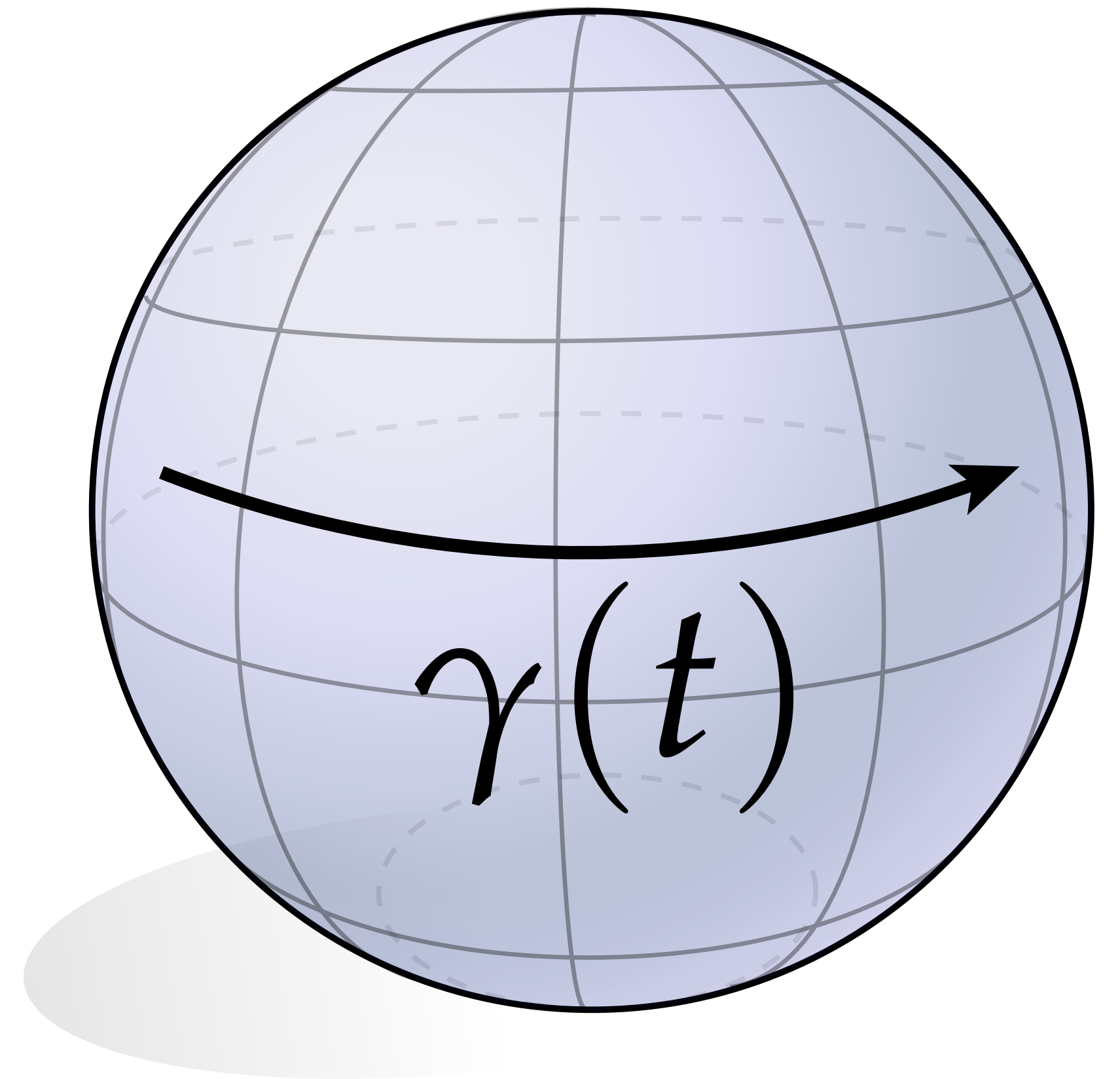
Geodesic Equation

The *covariant derivative* ∇ provides another characterization of geodesics:

$$\nabla_{\dot{\gamma}} \dot{\gamma} = 0$$

tangent to curve

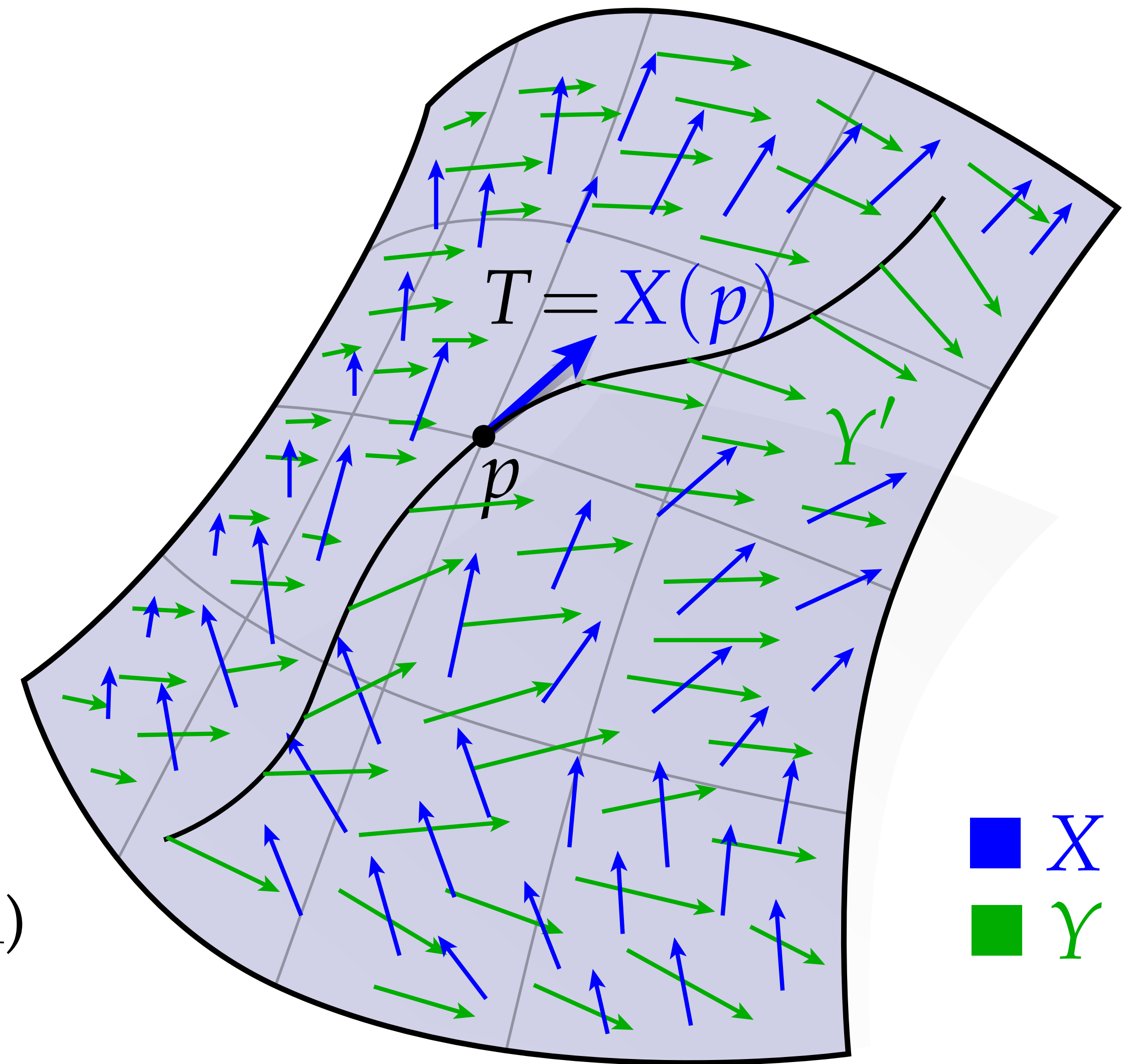
change along tangent direction



Intuition: no “in-plane turning” as we move along the curve.

Covariant Derivative—Extrinsic

- Suppose we want to measure how fast a vector field Y is changing along another vector field X at a point p
- Find a curve $\gamma(t)$ with tangent $X(p)$ at p
- Restrict Y to a vector field $Y'(t) := Y(\gamma(t))$
- Take the derivative dY'/dt
- Removing the normal component gives the *covariant derivative* $\nabla_X Y$ of Y along X
- Sound familiar?
 - not so different from how we defined *geodesic curvature* (change of T in B direction)
 - which explains geodesic equation $\nabla_{\dot{\gamma}} \dot{\gamma} = 0$



Key idea: covariant derivative gives change in one vector field along another.

Covariant Derivative—Intrinsic Definition

- Since geodesics are intrinsic, can also define “straightness” using only the metric g
- For any function ϕ , tangent vector fields X, Y, Z , operator ∇ uniquely determined by

$$\begin{aligned}\nabla_Z(X + Y) &= \nabla_Z X + \nabla_Z Y \\ \nabla_{X+Y} Z &= \nabla_X Z + \nabla_Y Z \\ \nabla_{\phi X} Y &= \phi \nabla_X Y \\ \nabla_X(\phi Y) &= (D_X \phi)Y + \phi \nabla_X Y\end{aligned}$$

(linearity)

(product rule)

$$D_Z g(X, Y) = g(\nabla_Z X, Y) + g(X, \nabla_Z Y)$$
$$\nabla_X Y - \nabla_Y X = [X, Y]$$

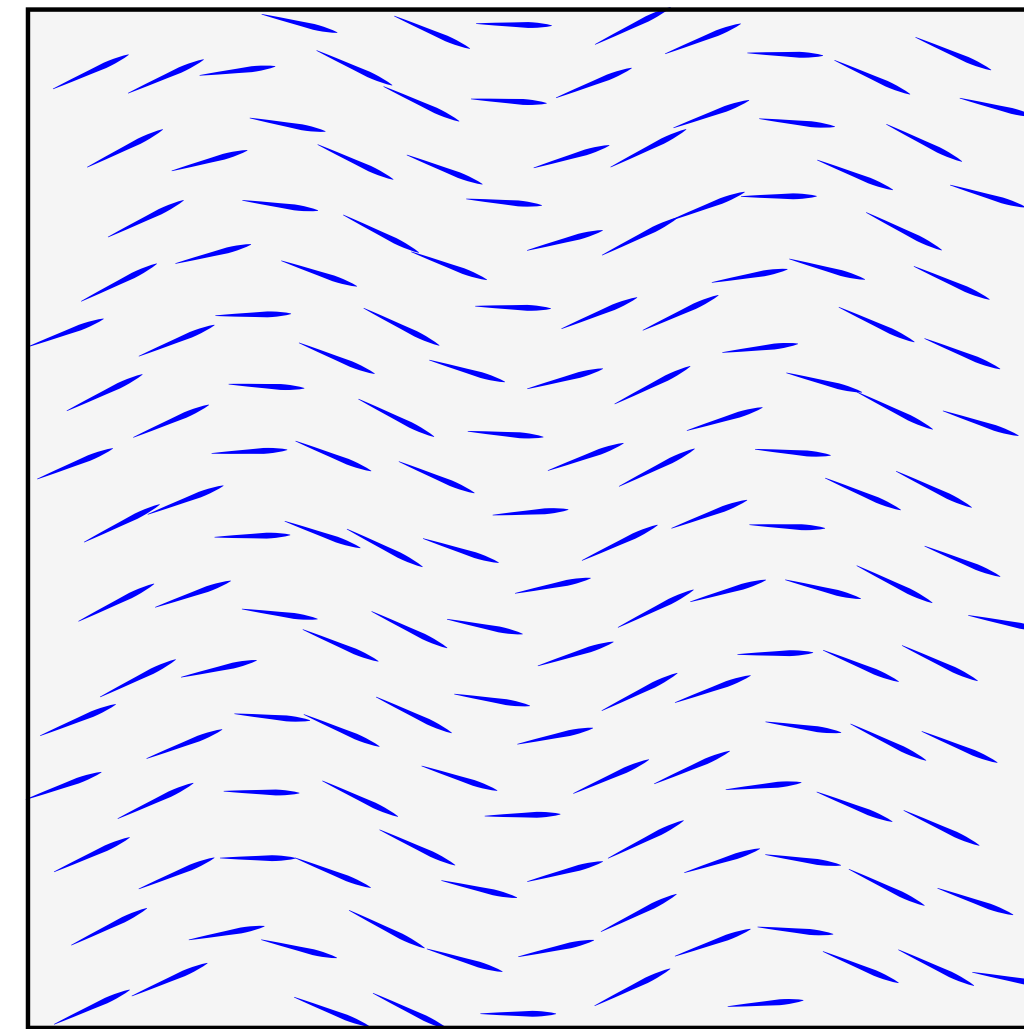
“compatible w/ metric”

“torsion free”

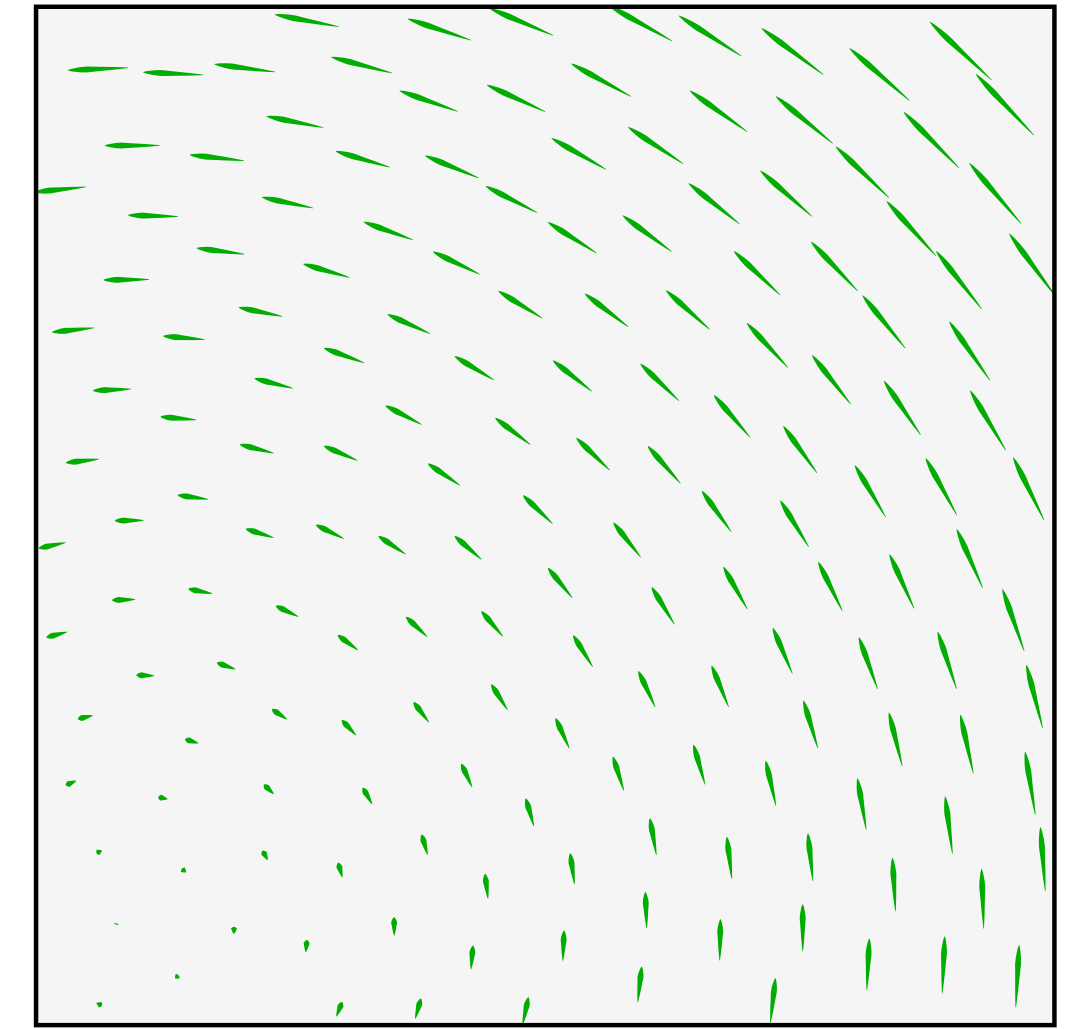
Lie Bracket of Vector Fields

- The *Lie bracket* $[X, Y]$ measures failure of flows along two vector fields X, Y to commute
- Starting at any point p , follow X for time $\tau > 0$, then Y , then $-X$, then $-Y$ to arrive at a point q
- Lie bracket at p is vector given by limit of $(q-p)/\tau$ as $\tau \rightarrow 0$
- For vector fields expressed in local coordinates $u_1, \dots, u_n$, can write as

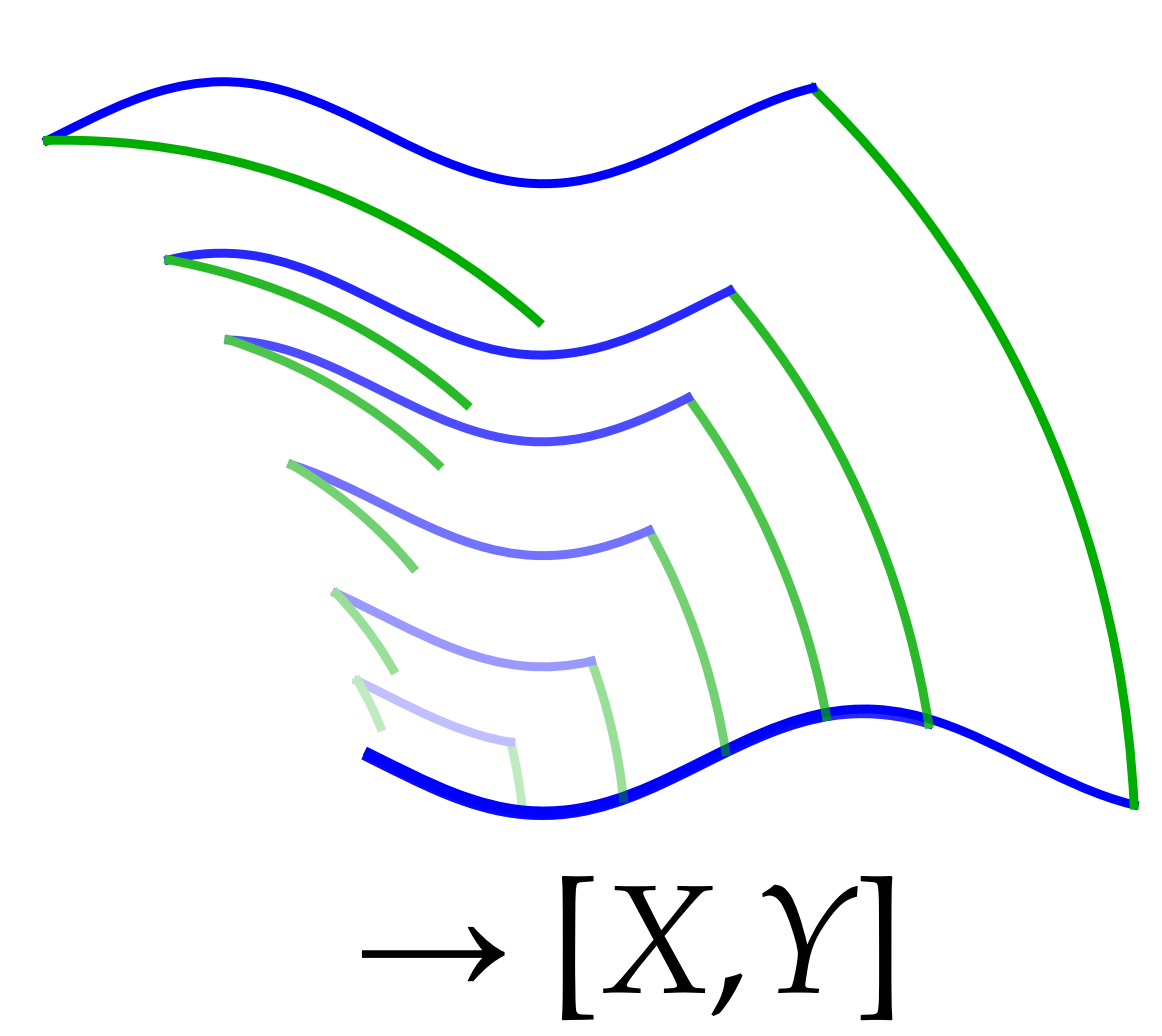
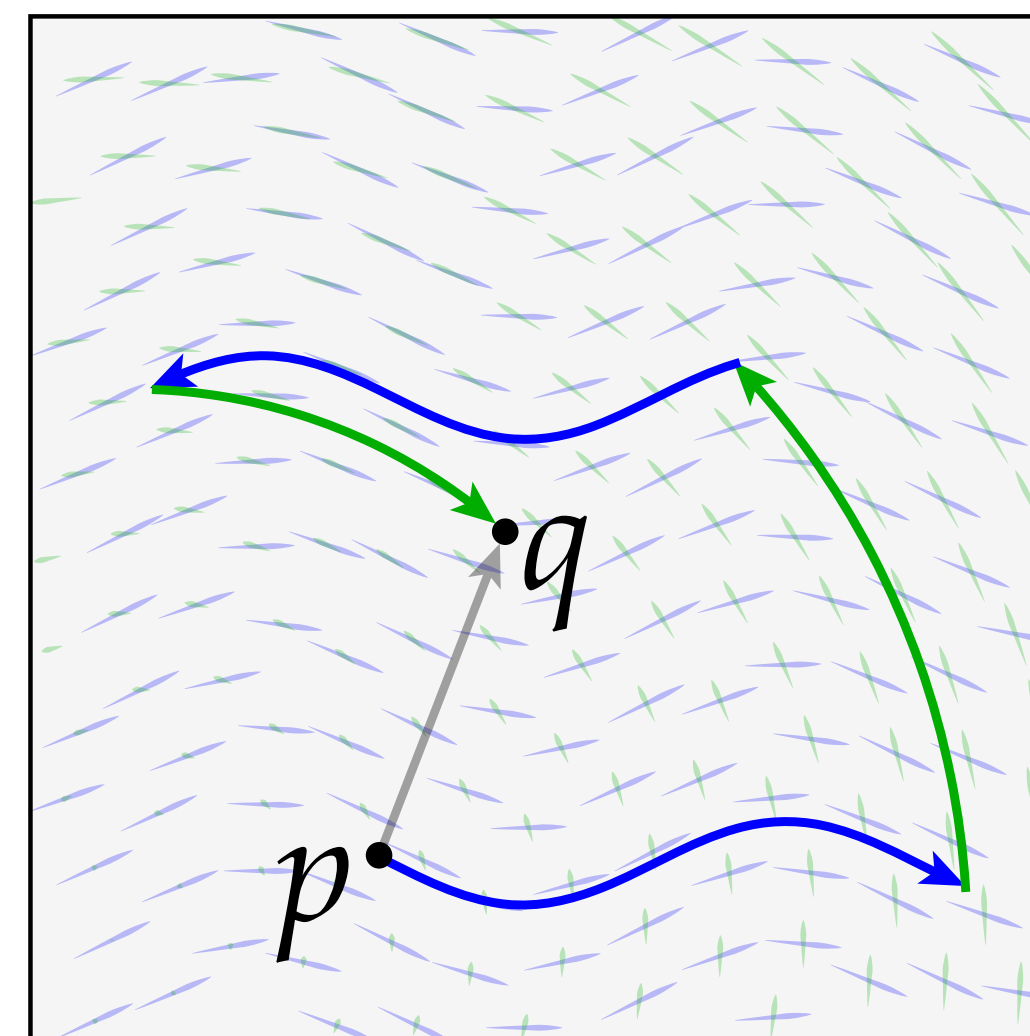
$$[X, Y] = \sum_{i,j=1}^n \left(X^j \frac{\partial}{\partial u_j} Y^i - Y^j \frac{\partial}{\partial u_j} X^i \right) \frac{\partial}{\partial u^i}$$



X



Y



$\rightarrow [X, Y]$

Covariant Derivative from Metric

Claim. Covariant derivative is uniquely determined by the Riemannian metric g .

Proof. For any three vector fields U, V, W , we have

$$D_U g(V, W) = g(\nabla_U V, W) + g(V, \nabla_U W) \quad (1)$$

$$D_V g(W, U) = g(\nabla_V W, U) + g(W, \nabla_V U) \quad (2)$$

$$D_W g(U, V) = g(\nabla_W U, V) + g(U, \nabla_W V) \quad (3)$$

By symmetry and bilinearity of the metric g , adding (1) and (2) and subtracting (3) gives

$$\begin{aligned} D_U g(V, W) + D_V g(W, U) - D_W g(U, V) &= \\ g(\nabla_U V + \nabla_V U) + g([U, W], V) + g([V, W], U) &= \\ 2g(\nabla_V U, W) + g([U, V], W) + g([V, W], U) + g([U, W], V). \end{aligned}$$

Hence,

$$g(\nabla_V U, W) = \frac{1}{2} (D_U g(V, W) + D_V g(W, U) - D_W g(U, V) - g([U, V], W) - g([V, W], U) - g([U, W], V)).$$

Key observation: can solve for covariant derivative in terms of data we know (metric g).

Christoffel Symbols

- Let $X_1, \dots, X_n$ be our usual basis vector fields (in local coordinates)
- *Christoffel symbols* tell us how to differentiate one basis along another: $\nabla_{X_j} X_i = \Gamma_{ij}^k X_k$
- By linearity, we then know how to take *any* covariant derivative

Recall the expression

$$g(\nabla_V U, W) = \frac{1}{2} (D_U g(V, W) + D_V g(W, U) - D_W g(U, V) - g([U, V], W) - g([V, W], U) - g([U, W], V)).$$

Since $[X_i, X_j] = 0$ for any two coordinate vector fields, we get

$$2g(\nabla_{X_k} X_i, X_j) = D_{X_i} g(X_k, X_j) + D_{X_k} g(X_j, X_i) - D_{X_j} g(X_i, X_k).$$

In terms of Christoffel symbols, the left-hand side is

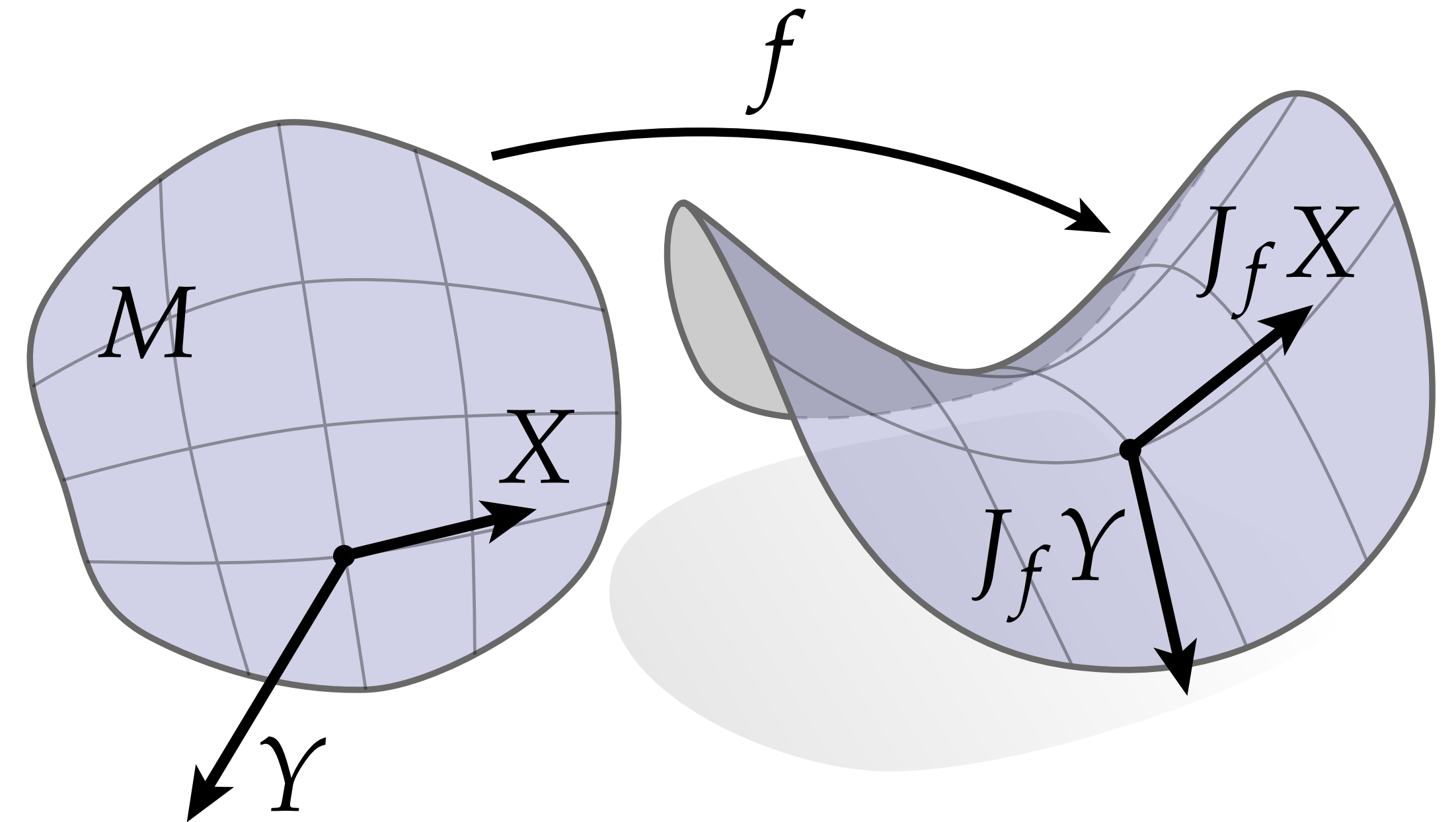
$$2g(\Gamma_{ik}^p X_p, X_j) = 2\Gamma_{ik}^p g(X_p, X_j) = 2\Gamma_{ik}^p g_{pj}$$

and we can write the right-hand side as $g_{kj,i} + g_{ji,k} - g_{ik,j}$.

Hence, our final expression for the Christoffel symbols is $\Gamma_{ik}^p = \frac{1}{2} g^{pj} (g_{ij,k} + g_{jk,i} - g_{ki,j})$

Solving the Geodesic Equation

- Can use Christoffel symbols to numerically compute geodesics on smooth surfaces
- Given surface $f : M \rightarrow \mathbb{R}^3$
 - write out Jacobian J_f
 - write out metric $g = J_f^\top J_f$ and its inverse g^{ij}
 - write out Christoffel symbols Γ
 - express geodesic equation via Γ
- From here, can use any standard numerical integrator (e.g., Runge-Kutta) to step an initial position/direction forward in “time”



$$\Gamma_{ik}^p = \frac{1}{2} g^{pj} (g_{ij,k} + g_{jk,i} - g_{ki,j})$$

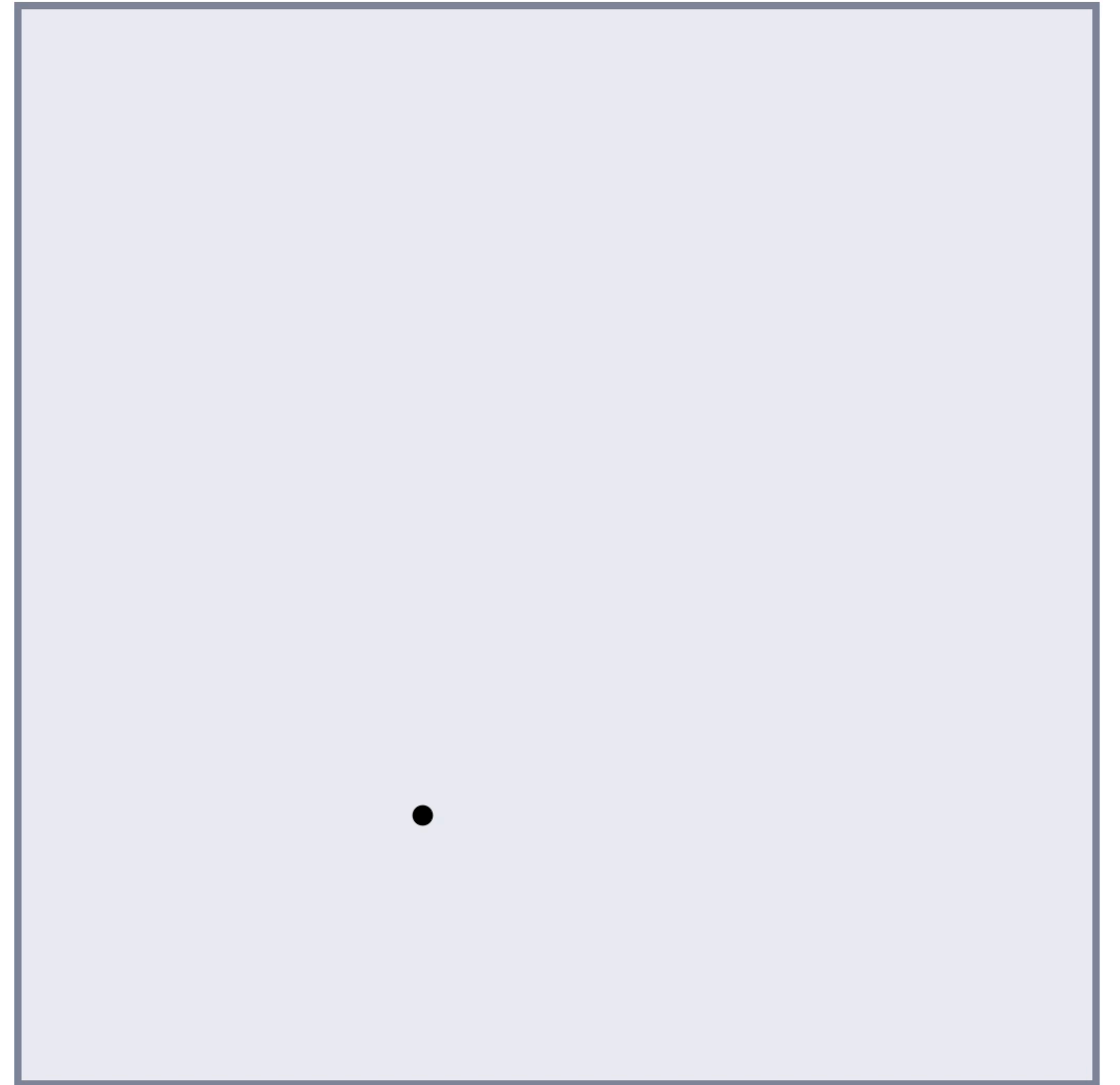
$$\nabla_{\dot{\gamma}} \dot{\gamma} = 0 \qquad \nabla_{X_j} X_i = \Gamma_{ij}^k X_k$$

$$\Rightarrow \boxed{\ddot{\gamma}^k + \Gamma_{ij}^k \dot{\gamma}^i \dot{\gamma}^j = 0}$$

Solving the Geodesic Equation



- Apply f to resulting curve in parameter domain to get a geodesic on the surface

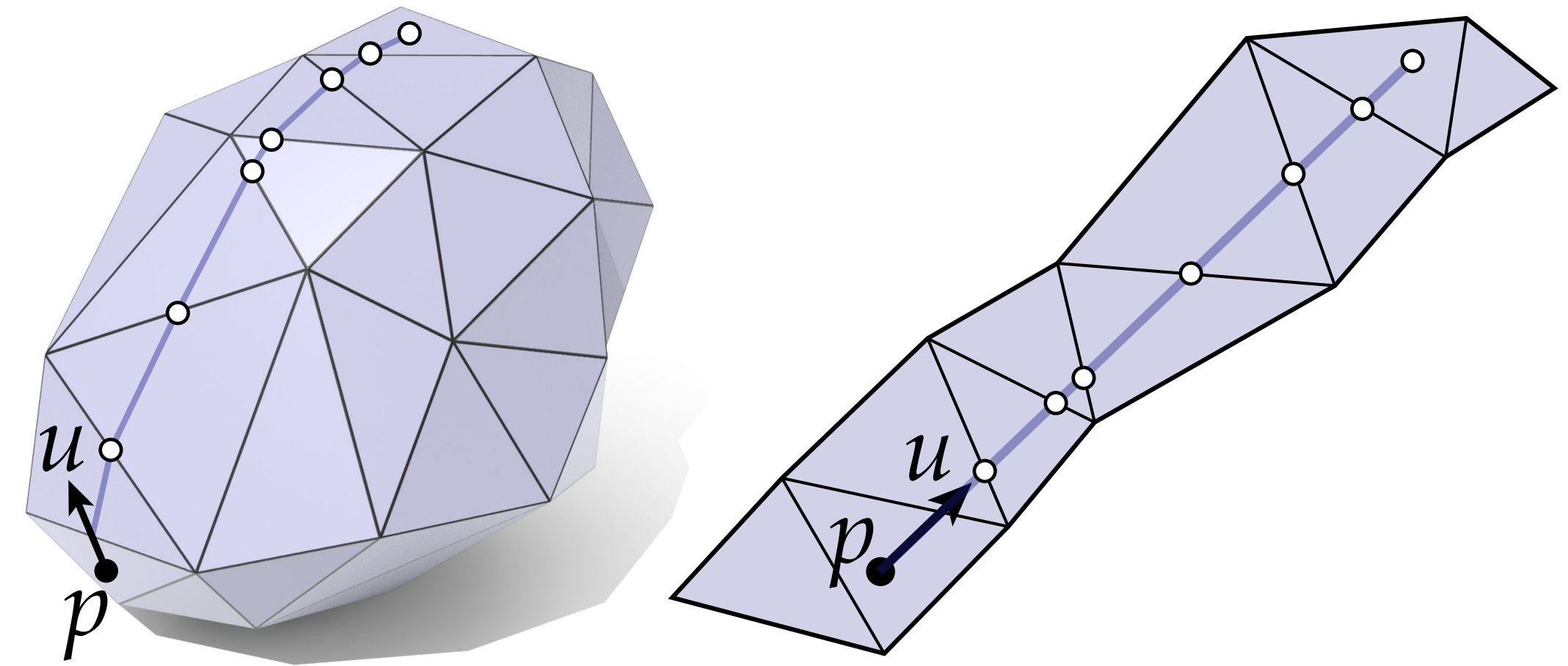


$$\Rightarrow \boxed{\ddot{\gamma}^k + \Gamma_{ij}^k \dot{\gamma}^i \dot{\gamma}^j = 0}$$

Computing Geodesics on a Parametrized Surface

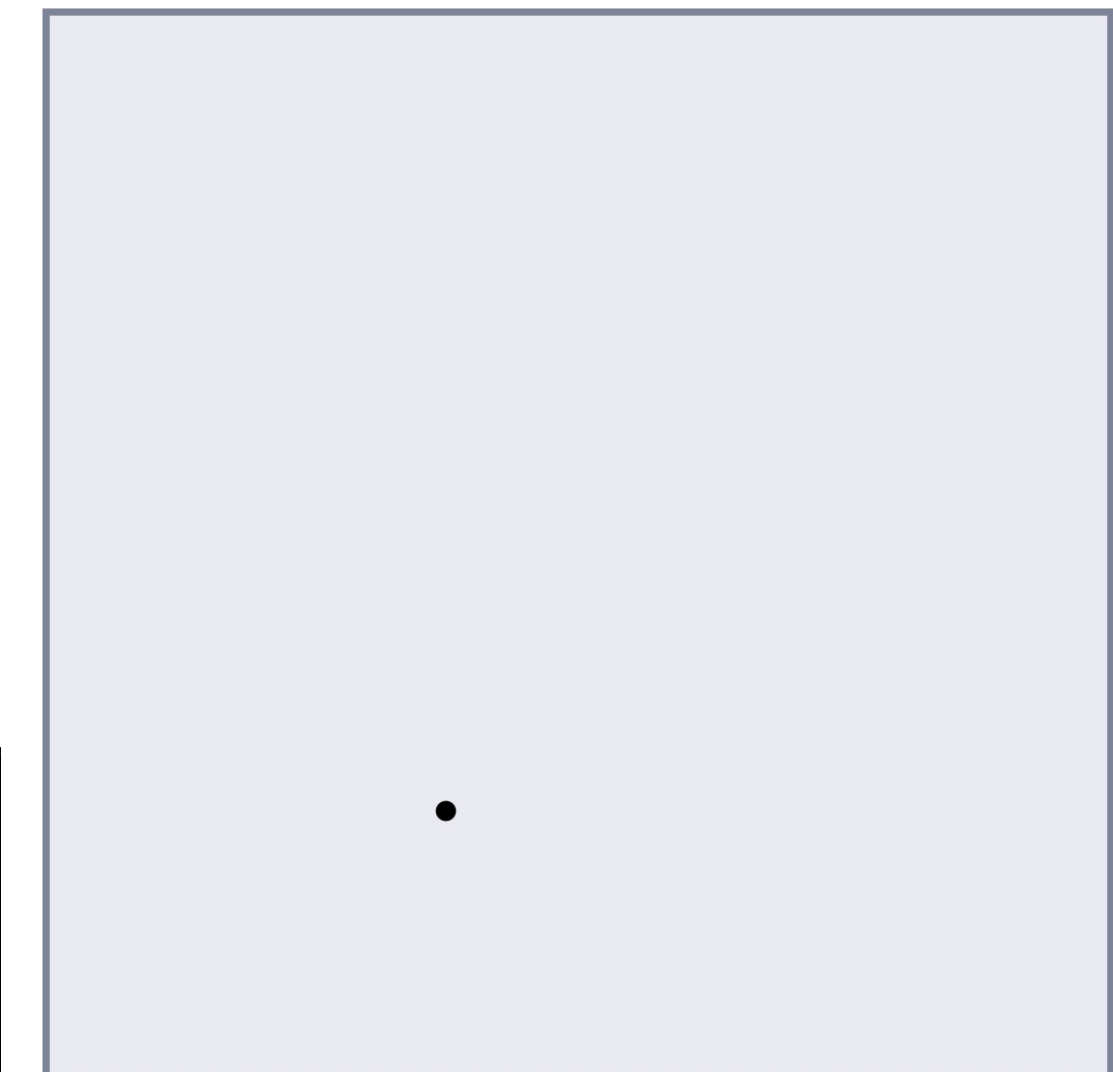
Now have two ways to solve initial value problem for a smooth parameterized surface f :

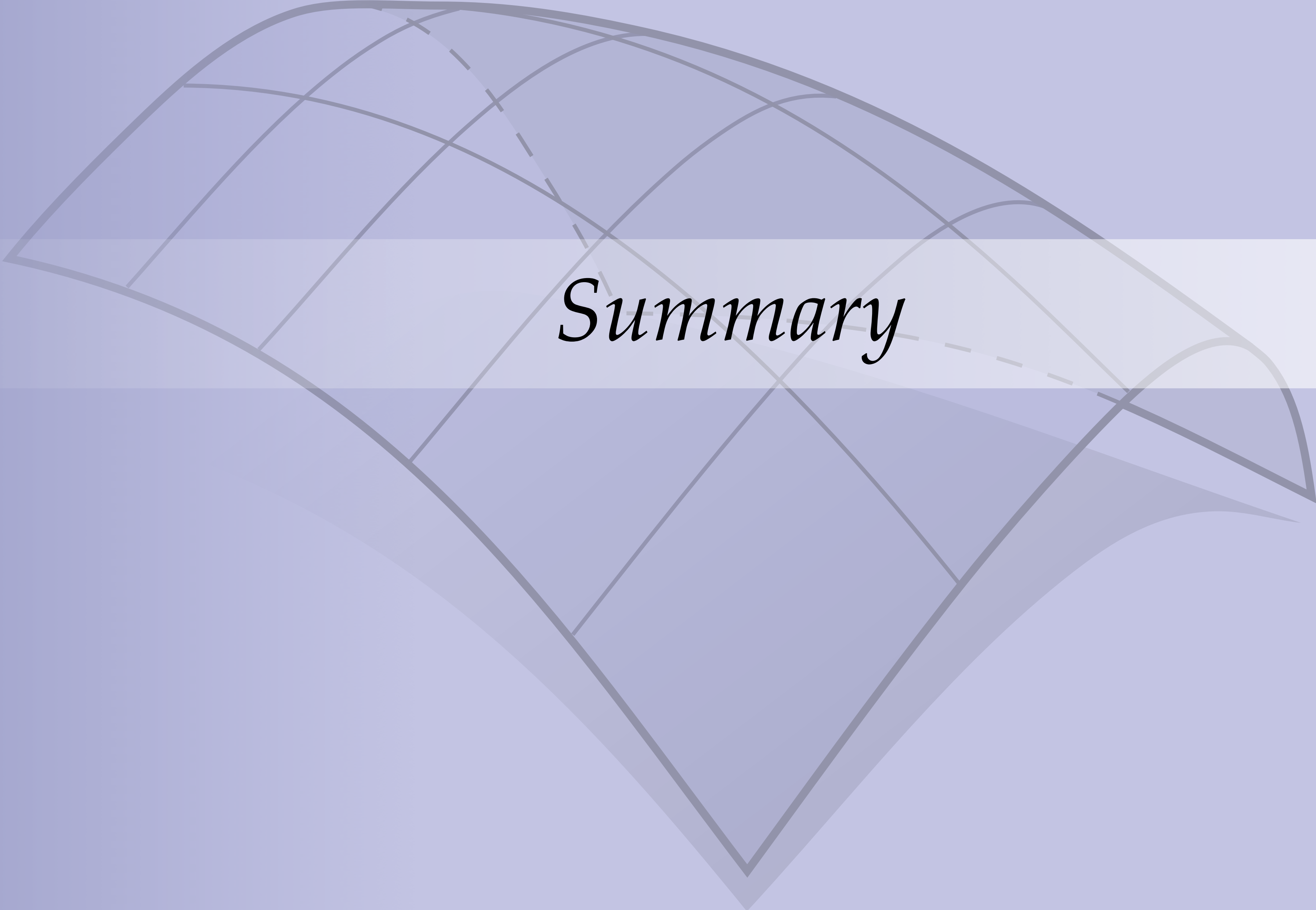
- Discretization
 - triangulate the surface f
 - trace rays along discrete surface
 - ODE integration
 - write metric g in terms of f
 - write Christoffel symbols Γ in terms of g
 - solve geodesic equation via ODE solver
- Q:** What are the pros/cons?
- speed, memory, accuracy, *simplicity*...
 - generality (smooth *and* discrete)



$$g = J_f^\top J_f$$
$$\underbrace{\frac{1}{2} g^{pj} (g_{ij,k} + g_{jk,i} - g_{ki,j})}_{\Gamma_{ik}^p}$$

$$\ddot{\gamma}^k + \Gamma_{ij}^k \dot{\gamma}^i \dot{\gamma}^j = 0$$

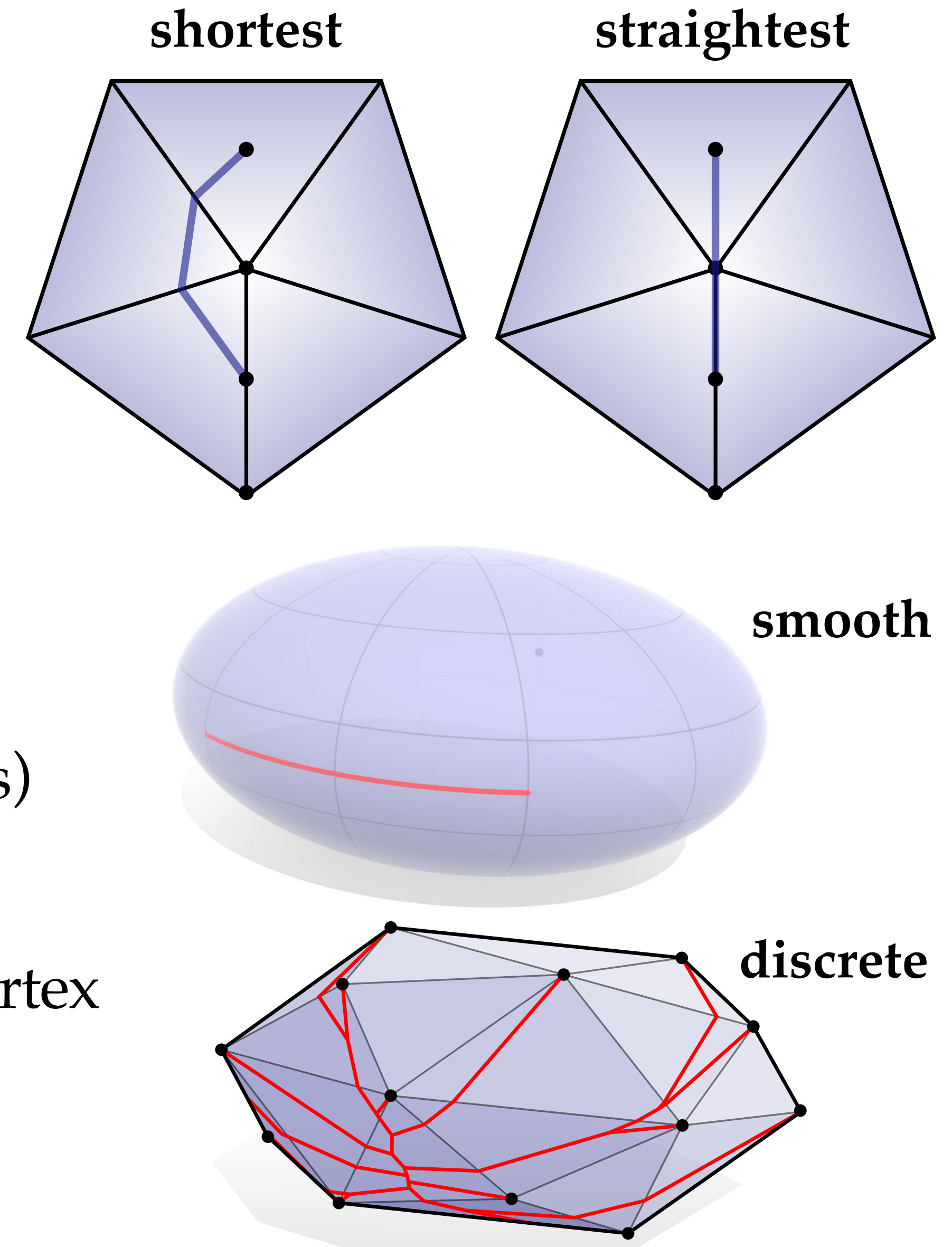




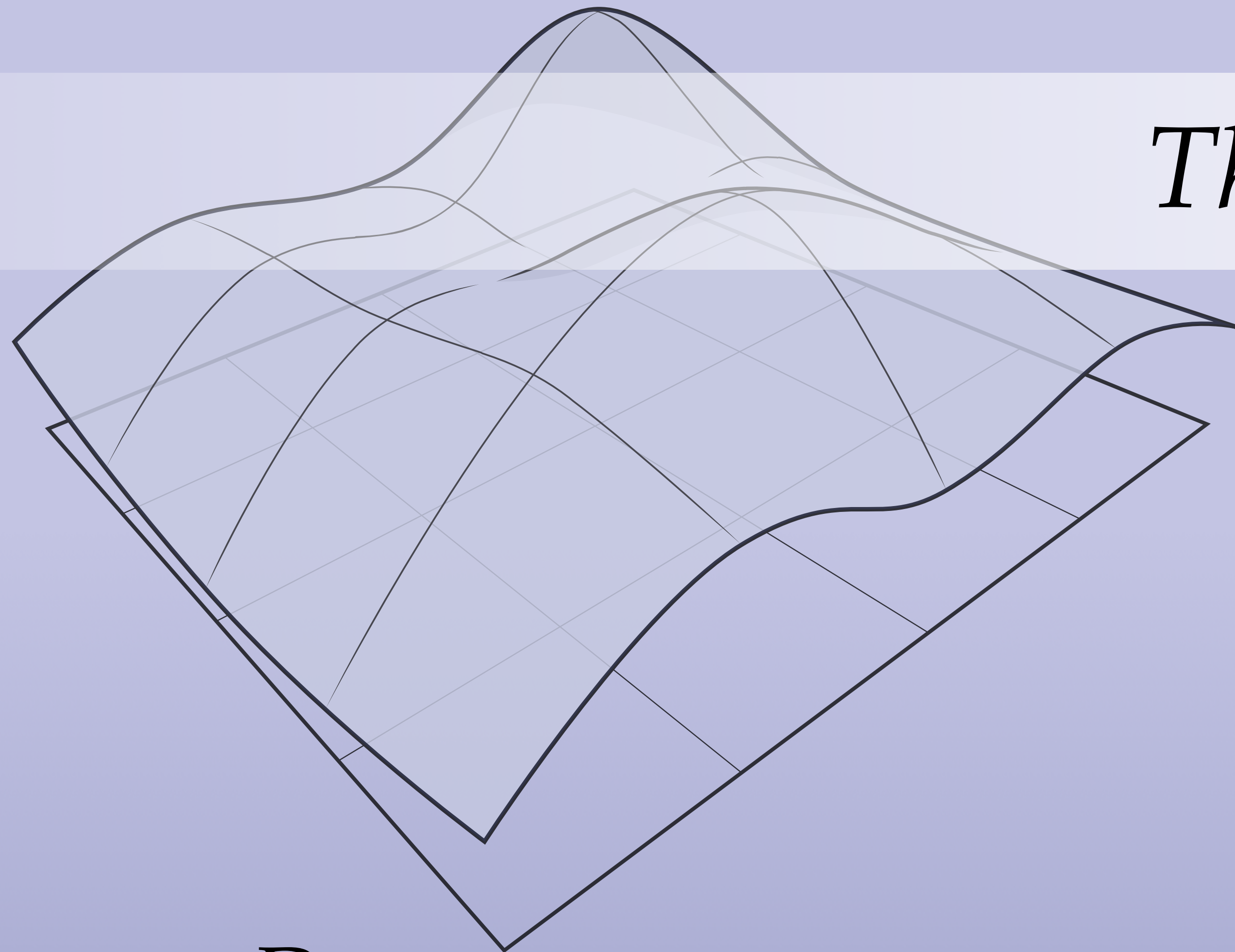
Summary

Geodesics — Shortest vs. Straightest, Smooth vs. Discrete

- In smooth setting, several equivalent characterizations:
 - shortest (harmonic)
 - straightest (zero curvature, zero acceleration)
- In discrete setting, characterizations no longer agree!
 - **shortest** natural for boundary value problem
 - **straightest** natural for initial value problem
 - *convex*: shortest paths are straightest (but not vice versa)
 - *nonconvex*: shortest may not even be straightest! (saddles)
- *Neither* definition faithfully captures all smooth behavior:
 - (shortest) cut locus / medial axis touches *every* convex vertex
 - (straightest) exponential map is not surjective
- Use the right tool for the job (*and look for other definitions!*)



Thanks!



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